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Registration-Free Learnable Multi-View Capture of Faces in Dense Semantic Correspondence

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Figure 1. **MOCHI**: Registration-free multi-view face capture in dense correspondence. Given calibrated multi-view images MOCHI predicts a canonical FLAME-topology mesh *without needing precomputed registrations during training*. When a target scan is available at test-time, an optional brief TTO of the model sharpens details and *outperforms classical, manually labor-heavy registration pipelines*.

Abstract

*Recent frameworks like ToFu and TEMPEH provide an automated alternative to classical registration pipelines by predicting 3D meshes in dense semantic correspondence directly from calibrated multi-view images. However, these learning-based methods rely on the slow, manual registration pipelines they aim to replace for their training supervision. We overcome this limitation with MOCHI (**M**ulti-view **O**ptimizable **C**orrespondence of **H**eads from **I**mages), a multi-view 3D face prediction framework trained without requiring registered training data. MOCHI eliminates the registration data dependency by enforcing topological consistency through a pseudo-linear inverse kinematic solver. Semantic alignment is guided by dense keypoints from a 2D landmark predictor trained exclusively on synthetic data. Our analysis further reveals that standard point-to-surface distances induce training instabilities and visual artifacts in registration-free settings. We propose pointmap- and normal-based losses instead, which provide smoother gradients and superior reconstruction fidelity. Finally, we introduce a test-time optimization scheme that refines network weights over a few dozen iterations. This approach bridges the gap between feed-forward efficiency and iterative optimization precision, allowing MOCHI to outperform traditional labor-intensive pipelines in both reconstruction accuracy and visual quality. Code and model are public at:* https://filby89.github.io/mochi.

1. Introduction

3D face reconstruction in dense correspondence – that is, producing meshes with a consistent topology across identities and expressions at the highest possible quality – is a central problem in computer vision. It has direct impact on a number of industries such as entertainment (VFX in movies, games), future forms of communication (teleconferencing in AR, VR, Metaverse), healthcare (such as cosmetic surgery pre-visualization), and perhaps future 3D embodied AI assistants.

Ideally, a face reconstruction method should not only re-

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无注册可学习的多视角人脸密集语义对应采集

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图1. MOCHI：密集对应下的免配准多视角人脸捕捉。给定已标定的多视角图像MOCH I预测一个规范的FLAME拓扑网格*without needing precomputed registrations during training*。当目标扫描可用时在测试时，对模型进行可选的简短TTO可锐化细节和*outperforms classical, manually labor-heavy registration pipelines*。

摘要

*Recent frameworks like ToFu and TEMPEH provide an automated alternative to classical registration pipelines by predicting 3D meshes in dense semantic correspondence directly from calibrated multi-view images. However, these learning-based methods rely on the slow, manual registration pipelines they aim to replace for their training supervision. We overcome this limitation with MOCHI (**M**ulti-view **O**ptimizable **C**orrespondence of **H**eads from **I**mages), a multi-view 3D face prediction framework trained without requiring registered training data. MOCHI eliminates the registration data dependency by enforcing topological consistency through a pseudo-linear inverse kinematic solver. Semantic alignment is guided by dense keypoints from a 2D landmark predictor trained exclusively on synthetic data. Our analysis further reveals that standard point-to-surface distances induce training instabilities and visual artifacts in registration-free settings. We propose pointmap- and normal-based losses instead, which provide smoother gradients and superior reconstruction fidelity. Finally, we introduce a test-time optimization scheme that refines network weights over a few dozen iterations. This approach bridges the gap between feed-forward efficiency and iterative optimization precision, allowing MOCHI to outperform traditional labor-intensive pipelines in both reconstruction accuracy and visual quality. Code and model are public at:* https://filby89.github.io/mochi.

1. 引言

三维人脸稠密对应重建——即在身份和表情之间以最高质量生成具有一致拓扑结构的网格——是计算机视觉中的一个核心问题。它对多个行业产生直接影响，例如娱乐业（电影中的视觉特效、游戏）、未来的通信形式（增强现实、虚拟现实、元宇宙中的远程会议）、医疗保健（如整容手术预可视化），以及未来可能的3D具身AI助手。

理想情况下，人脸重建方法不仅应重

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cover accurate geometry but also enforce a shared topology across individuals and frames, enabling semantic consistency and compatibility with downstream tasks such as animation and editing. Despite tremendous progress in face modeling and reconstruction [21, 77], the highest-fidelity systems still rely on multi-view stereo (MVS) capture [26, 67] followed by non-rigid registration to bring scans into a predefined topology [11, 48, 59]. This registration step requires iterative optimization and often manual verification or correction by human experts, making it one of the major bottlenecks for scalable 3D face capture.

Recent learning-based systems aim to simplify this process by predicting meshes in a fixed topology directly from calibrated multi-view images [9, 46, 49] (see Fig. 2). While such methods greatly reduce processing time, they still *require precomputed registrations during training*, and preparing such data remains time-consuming and labor-intensive, and a classical registration pipeline must be in place to provide the training supervision.

To remove the dependency on precomputed registrations for training, we present MOCHI, a *registration-free* approach for multi-view 3D face reconstruction in dense correspondence which allows training directly on raw scans. First, a pseudo-linear inverse kinematic solver [70] recovers 3DMM parameters from the network's vertex outputs. These parameters are then passed through the parametric model to produce a topology-consistent mesh, which is used to regularize the network's free-form predictions via vertex- and edge-level losses. Second, we observe that the traditionally used loss to train with scans, i.e., the point-to-surface distance, produces noisy gradients due to discrete closest point assignments thus often leading to self-intersections and other artifacts if the mesh is not already tightly aligned to the scan. We instead swap this out for differentiable rendering losses: *pointmap* and *surface-normal* losses [81]. These losses provide smoother gradients and train the model more reliably. Finally, because direct geometric errors often fail to capture semantically salient perceptual effects, we additionally supervise with 2D landmarks predicted by a model trained purely on synthetic data. At inference, we also introduce a *test-time optimization* (TTO) stage that performs a brief per-instance optimization of the model parameters, leading to registrations *more accurate* than the classical manually intensive registration pipeline. As a result, MOCHI can register a new multi-view dataset from scratch, using only the raw scans and calibrated images, effectively minimizing the manual effort needed (Fig. 2).

Our contributions can be summarized as follows: (1) To our knowledge, the first dense-correspondence multi-view face reconstruction system that does not require precomputed registrations for training, yet surpasses registration-requiring methods such as TEMPEH [9] in reconstruction accuracy. (2) A stable training objective based on pointmap and surface-normal losses, reducing artifacts and improving model training compared to scan-to-mesh supervision. (3) A lightweight test-time optimization (TTO) of our base model, which when coupled with an existing scan outperforms even classical manually-intensive registration pipelines.

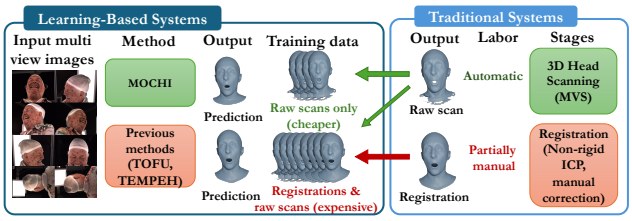

Figure 2. *Traditional capturing systems* (right) rely on a multi-stage approach involving MVS to produce a head scan, followed by registration that requires manual supervision. Recent *learning-based systems* (left) simplify the workflow but still require GT registrations from traditional systems for training (red arrow). *MOCHI* relies only on the fully automatic parts (green arrow), outperforming previous learning-based methods despite removing the need for expensive manual registrations. When coupled with TTO, MOCHI surpasses even the classic manual registration pipeline.

2. Related Work

Recovering 3D faces from images has advanced dramatically since the seminal work of Blanz and Vetter [7]. We review the most relevant lines of work below.

Scan Registration. Traditional high-fidelity face capture begins with obtaining a 3D scan, most commonly via multi-view stereo (MVS) [22, 26, 34, 43, 80], though laser scanning [44], photometric stereo [31, 56], and learning-based methods [38, 42, 81, 83, 92] are also used; a comprehensive survey [82] covers these in detail. The resulting scans have unstructured topology and may contain holes or artifacts in difficult regions (e.g. lips, eyes, facial hair), and must therefore be registered onto a common topology to be useful in production. A common approach is to first fit a 3DMM to the scan [1, 8, 47], providing a coarse initialization that is then refined with non-rigid ICP [6] to capture finer detail [10, 12, 15, 30, 45, 48, 58, 59, 65, 87, 91]. Other methods skip the model initialization and apply non-rigid ICP directly to deform a template to the scan [4, 5, 13]. Some works collapse scanning and registration into a single joint stage [27, 63]. Learning-based approaches train neural networks to map unstructured raw scans directly to a shared template topology, through mechanisms such as surface-to-surface translation [3], predicting point-wise displacements [50], or learning continuous implicit deformation fields [95, 96]. These are faster than iterative optimization but remain prone to failures on noisy inputs. Overall, the above registration pipelines yield excellent fidelity but

不仅覆盖精确的几何形状，还强制跨个体和帧共享拓扑，从而实现语义一致性以及与动画和编辑等下游任务的兼容性。尽管面部建模与重建[21, 77]取得了巨大进展，但最高保真度的系统仍依赖于多视角立体（MVS）采集[26, 67]，随后通过非刚性配准将扫描数据转换到预定义拓扑[11, 48, 59]。这一配准步骤需要迭代优化，且通常需要人工专家进行手动验证或修正，使其成为可扩展3D面部采集的主要瓶颈之一。

近年来，基于学习的系统旨在通过直接从校准的多视图图像预测固定拓扑的网格来简化这一过程[9, 46, 49]（见图2）。尽管此类方法大幅减少了处理时间，它们仍然*require precomputed registrations during training*，且准备此类数据仍然耗时费力，并且必须建立经典的配准流程以提供训练监督。

为了消除训练时对预计算配准的依赖，我们提出了MOCHI，一种用于稠密对应多视图三维人脸重建的*registration-free*方法，该方法允许直接在原始扫描数据上进行训练。首先，一个伪线性逆运动学求解器[70]从网络的顶点输出中恢复3DMM参数。随后，这些参数通过参数化模型生成拓扑一致的网格，并利用顶点级和边级损失对网络的自由形式预测进行正则化。其次，我们观察到传统上用于扫描训练的损失，即点到表面距离，由于离散最近点分配而产生噪声梯度，若网格尚未与扫描紧密对齐，往往导致自交和其他伪影。我们转而采用可微渲染损失：*pointmap*和*surface-normal*损失[81]。这些损失提供更平滑的梯度，使模型训练更加可靠。最后，由于直接几何误差往往无法捕捉语义上显著的感知效果，我们还额外使用一个仅在合成数据上训练的模型所预测的二维地标进行监督。在推理阶段，我们还引入了一个*test-time optimization* (TTO)阶段，对模型参数进行简短的逐实例优化，从而得到比传统人工密集型配准流程更*more accurate*的配准结果。因此，MOCHI能够仅使用原始扫描和标定图像从头开始配准新的多视图数据集，有效减少所需的人工工作量（图2）。

我们的贡献可总结如下：（1）据我们所知，这是首个无需预计算配准即可训练、且在重建精度上超越TEMPEH [9]等需配准方法的稠密对应多视角人脸重建系统。

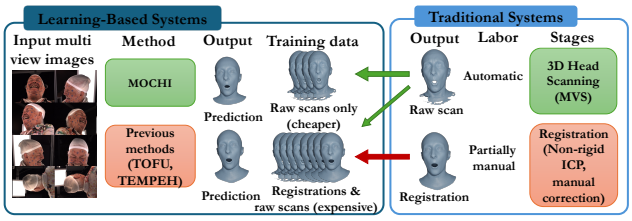

图2. *Traditional capturing systems* (右)依赖涉及MVS的多阶段方法来生成头部扫描，随后进行需要人工监督的配准。最近的*learning- based systems* (左)简化了工作流程，但仍需要来自传统系统的GT配准用于训练（红色箭头）。*MOCHI*仅依赖全自动部分（绿色箭头），尽管去除了昂贵的人工配准需求，仍优于先前的基于学习的方法。当与TTO结合时，MOCHI甚至超越了经典的人工配准流程。

（2）一种基于点图和表面法向损失的稳定训练目标，与扫描到网格的监督方式相比，减少了伪影并改善了模型训练。（3）对基础模型进行轻量级测试时优化（TTO），当与现有扫描结合时，其性能甚至优于经典的手动密集型配准流程。

2. 相关工作

自Blanz和Vetter的开创性工作[7]以来，从图像中恢复3D人脸的技术取得了显著进展。下面我们回顾最相关的几项研究工作。

扫描配准。传统高保真面部捕捉首先需要获取3D扫描，最常用的是多视角立体视觉（MVS）[22, 26, 34, 43, 80]，但也使用激光扫描[44]、光度立体视觉[31, 56]和基于学习的方法[38, 42, 81, 83, 92]；一篇综合性综述[82]对此进行了详细讨论。生成的扫描具有非结构化拓扑，在困难区域（如嘴唇、眼睛、面部毛发）可能包含孔洞或伪影，因此必须配准到通用拓扑上才能在制作中发挥作用。一种常见的方法是先将3DMM拟合到扫描上[1, 8, 47]，提供粗略的初始化，然后通过非刚性ICP[6]进行细化以捕捉更精细的细节[10, 12, 15, 30, 45, 48, 58, 59, 65, 87, 91]。其他方法跳过模型初始化，直接应用非刚性ICP将模板变形到扫描上[4, 5, 13]。一些工作将扫描和配准合并为单个联合阶段[27, 63]。基于学习的方法训练神经网络，通过表面到表面的转换[3]、预测逐点位移[50]或学习连续隐式变形场[95, 96]等机制，将非结构化原始扫描直接映射到共享模板拓扑。这些方法比迭代优化更快，但在噪声输入上仍然容易失败。总体而言，上述配准流程产生了出色的保真度，但

require heavy computation and substantial manual cleanup (e.g. resolving teeth artifacts, fixing failures on extreme expressions, and tuning hyperparameters per subject). In contrast, MOCHI removes the need for any registration pipeline by learning directly from raw scans.

Learnable Multi-View Mesh Prediction. A recent line of deep-learning-based methods [9, 46, 49, 51] aims to collapse the multi-stage, partly manual traditional pipeline into a single neural network that can be run without manual correction (see Fig. 2). ToFu [49] pioneered this approach, using a volumetric deep feature map to predict vertices of a predefined template from multi-view calibrated images. ReFA [51] instead uses deep features to refine 3D position and normal maps in UV space. TEMPEH [9] builds upon ToFu by considering surface properties and visibility for multi-view feature fusion, and is supervised mostly with unstructured scans, but still needs at least coarse registrations for regularization. GRAPE [46] builds upon TEMPEH, proposing a visual hull and visibility-aware 3D feature aggregation to generalize to novel camera rigs. Xiao et al. [89] learn an implicit function to recover detailed facial geometry from multi-view images, supervised with GT displacement maps derived from registered templates. All of the above methods require GT registrations and hence still need the traditional pipeline to create them (see Fig. 2). With MOCHI, we go one step further: we employ a pseudo-linear inverse kinematic solver [70] to estimate FLAME [48] parameters, outputting a mesh that is used as a regularizer. This step, combined with our differentiable rendering losses and semantic guidance from dense 2D landmarks, *effectively removes the need for GT registrations*. Note that while dense landmarks have been used previously in 3D face reconstruction [74, 85], in our case they do not drive the geometric accuracy but act as a semantic regularizer for ambiguous regions such as eyes and lips. Finally, some methods sidestep explicit meshes altogether, using neural representations learned directly from images [2, 28, 33, 52, 53, 55, 90] to represent geometry via neural volumes, radiance fields, or Gaussian primitives. While these can produce high-fidelity renderings, they lack the editability and compatibility with standard graphics engines that explicit mesh-based representations, including ours, provide.

Image-based Face Reconstruction. A large body of work focuses on simplifying the capture setup from calibrated multi-view to uncalibrated single-view images; recent surveys [21, 77] give a comprehensive overview. Early works focused on optimization-based fitting of 3DMMs to images using "analysis-by-synthesis" [7, 14, 78, 79, 86]. Many deep-learning works repurposed this paradigm for self-supervised systems [17, 24, 25, 36, 60, 62, 66, 69, 75, 76, 94], where a neural network maps image features onto 3DMM and scene parameters. A few recent works predict landmarks [74] or depth and UV maps [32] with post-hoc 3DMM fitting. Some methods bypass 3DMMs entirely and train on 3D GT to output meshes [18, 20, 23, 35, 41, 64, 68, 73, 84, 88, 93], voxels [39], or SDFs [57], but these unstructured outputs require post-hoc registration. Despite the progress, none of these methods reach the quality of studio-based capture.

3. Method

3.1. Preliminaries

FLAME. We adopt FLAME [48] as our parametric head model for representing facial geometry. It generates a mesh $\mathbf{V} \in \mathbb{R}^{n_v \times 3}$ with $n_v = 5023$ vertices as a function of shape $\boldsymbol{\beta}$, expression $\boldsymbol{\psi}$, and pose parameters $\boldsymbol{\theta}$. The mesh is computed by applying linear blend skinning (LBS) to a base template deformed by learned corrective bases:

$$\mathbf{V} = \mathrm{LBS}\left(\bar{\mathbf{V}} + \mathbf{B}_{\text{id}}\boldsymbol{\beta} + \mathbf{B}_{\text{exp}}\boldsymbol{\psi} + \mathbf{B}_{\text{pose}}(\boldsymbol{\theta}), \boldsymbol{\theta}, \mathcal{J}, \mathcal{W}\right),$$

where $\bar{\mathbf{V}}$ is the mean template, $\mathbf{B}_{\text{id}}$ and $\mathbf{B}_{\text{exp}}$ are identity and expression bases, and $\mathbf{B}_{\text{pose}}$ are pose-dependent corrective blendshapes. The LBS operator uses joint locations $\mathcal{J}$ and skinning weights $\mathcal{W}$ to apply articulated pose deformations.

3.2. Methodology

We follow the multi-view capture setting of TEMPEH [9] but train *without* registrations. Given K calibrated views $\{\mathcal{I}_i\}_{i=1}^{K}$ with corresponding cameras $\{C_i\}_{i=1}^{K}$, **MOCHI** predicts a canonical FLAME-topology mesh $M = (\mathbf{V}, \mathbf{T})$, where $\mathbf{V} \in \mathbb{R}^{n_v \times 3}$ are the n_v predicted vertex coordinates and $\mathbf{T}$ denotes the fixed set of triangular faces defining mesh connectivity. The mesh $\hat{M}$ should be spatially aligned to the captured scan geometry while maintaining the canonical FLAME topology.

To enable training directly from raw scans, we introduce three key extensions: (i) a pseudo-linear inverse-kinematics branch that recovers FLAME parameters from per-vertex predictions to *enforce* topology; (ii) dense 2D landmarks from a synthetic-data-trained detector that act as a semantic regularizer, constraining articulated regions; and (iii) fully differentiable point-map and surface-normal losses that replace scan-to-mesh distances, providing smoother gradients and improved reconstruction accuracy.

At inference, we further introduce an optional *test-time optimization* (TTO) stage that refines the predicted mesh $\hat{M}$. This refinement minimizes the same differentiable geometric losses used during training, allowing the network predictions to be locally adapted to the specific input scans without requiring explicit correspondences.

3.3. Dense Landmark Detector

Previous methods like TEMPEH [9] rely on manual registrations to supervise mesh prediction, since point-to-surface

需要大量计算和大量手动清理（例如解决牙齿伪影、修复极端表情下的失败，以及为每个受试者调整超参数）。相比之下，MOCHI通过直接从原始扫描中学习，省去了任何配准流程的需要。

可学习的多视图网格预测。近期一系列基于深度学习的方法[9, 46, 49, 51]旨在将多阶段、部分人工的传统流程压缩为单一神经网络，无需人工修正即可运行（见图2）。ToFu[49]开创了这一方法，利用体素深度特征图从多视图标定图像中预测预定义模板的顶点。ReFA[51]则使用深度特征在UV空间中细化3D位置和法线图。TEMPEH[9]在ToFu的基础上，考虑表面属性和可见性进行多视图特征融合，主要使用非结构化扫描进行监督，但仍需至少粗略的配准作为正则化。GRAPE[46]在TEMPEH的基础上，提出视觉外壳和可见性感知的3D特征聚合，以泛化到新的相机装置。Xiao等人[89]学习隐式函数，从多视图图像中恢复精细的面部几何，使用从配准模板导出的GT位移图进行监督。上述所有方法都需要GT配准，因此仍需传统流程来生成这些配准（见图2）。在MOCHI中，我们更进一步：采用伪线性逆运动学求解器[70]来估计FLAME[48]参数，输出网格作为正则化器。该步骤结合我们的可微渲染损失和来自密集2D地标的语义引导，*effectively removes the need for GT regis- trations*。需要注意的是，虽然密集地标此前已用于3D人脸重建[74, 85]，但在我们的方法中，它们并不驱动几何精度，而是作为眼睛和嘴唇等模糊区域的语义正则化器。最后，一些方法完全绕开显式网格，使用直接从图像学习的神经表示[2, 28, 33, 52, 53, 55, 90]，通过神经体积、辐射场或高斯原语来表示几何。虽然这些方法能产生高保真渲染，但它们缺乏显式基于网格的表示（包括我们的方法）所提供的可编辑性和与标准图形引擎的兼容性。

基于图像的人脸重建。大量工作致力于将采集设置从已标定的多视图简化为未标定的单视图图像；近期综述[21, 77]提供了全面概述。早期工作侧重于使用“分析合成”[7, 14, 78, 79, 86]将3DMM优化拟合到图像。许多深度学习工作将该范式重新用于自监督系统[17, 24, 25, 36, 60, 62, 66, 69, 75, 76, 94]，其中神经网络将图像特征映射到3DMM和场景参数。少数近期工作预测地标[74]或深度和UV图[32]配合事后3DMM拟合。一些方法完全绕过3DMM，直接在3D GT上训练以输出网格[18, 20, 23, 35, 41, 64, 68, 73, 84, 88, 93]、体素[39]或SDF[57]，但这些非结构化输出需要事后配准。尽管取得了进展，这些方法均未达到基于工作室捕捉的质量。

3. 方法

3.1. 预备知识

FLAME。我们采用FLAME [48]作为表示面部几何的参数化头部模型。它根据形状$\boldsymbol{\beta}$、表情$\boldsymbol{\psi}$和姿态参数$\boldsymbol{\theta}$生成具有$n_v = 5023$个顶点的网格$\mathbf{V} \in \mathbb{R}^{n_v \times 3}$。该网格通过对由学习到的修正基变形的基础模板应用线性混合蒙皮（LBS）计算得到：

$$\mathbf{V} = \mathrm{LBS}\left(\bar{\mathbf{V}} + \mathbf{B}_{\mathrm{id}}\boldsymbol{\beta} + \mathbf{B}_{\mathrm{exp}}\boldsymbol{\psi} + \mathbf{B}_{\mathrm{pose}}(\boldsymbol{\theta}), \boldsymbol{\theta}, \mathcal{J}, \mathcal{W}\right),$$

其中$\bar{\mathbf{V}}$为平均模板，$\mathbf{B}_{\mathrm{id}}$和$\mathbf{B}_{\mathrm{exp}}$分别为身份基和表情基，$\mathbf{B}_{\mathrm{pose}}$为姿态相关的修正混合形状。LBS算子利用关节点位置$\mathcal{J}$和蒙皮权重$\mathcal{W}$来施加关节姿态变形。

3.2. 方法

我们沿用TEMPEH [9]的多视角采集设置，但训练*without*配准。给定K个已标定视角$\{\mathcal{I}_i\}_{i=1}^{K}$及对应相机$\{C_i\}_{i=1}^{K}$，MOCHI预测一个规范FLAME拓扑网格$M = (\mathbf{V}, \mathbf{T})$，其中$\mathbf{V} \in \mathbb{R}^{n_v \times 3}$为$n_v$个预测顶点坐标，$\mathbf{T}$表示定义网格连接性的固定三角面集合。网格$\hat{M}$应在空间上与采集的扫描几何对齐，同时保持规范FLAME拓扑。

为了支持直接从原始扫描进行训练，我们引入了三项关键扩展：(i) 一个伪线性逆运动学分支，从逐顶点预测中恢复FLAME参数至*enforce*拓扑；(ii) 由合成数据训练的检测器生成的密集2D地标，作为语义正则化器，约束关节区域；(iii) 完全可微的点图和表面法线损失，替代扫描到网格的距离，提供更平滑的梯度并提升重建精度。

在推理阶段，我们进一步引入一个可选的*test-time optimization* (TTO)阶段，用于细化预测的网格$\hat{M}$。该细化过程最小化与训练时相同的可微几何损失，使网络预测能够针对特定输入扫描进行局部调整，而无需显式对应关系。

3.3. 密集地标检测器

先前的方法如TEMPEH [9]依赖手动配准来监督网格预测，因为点到表面

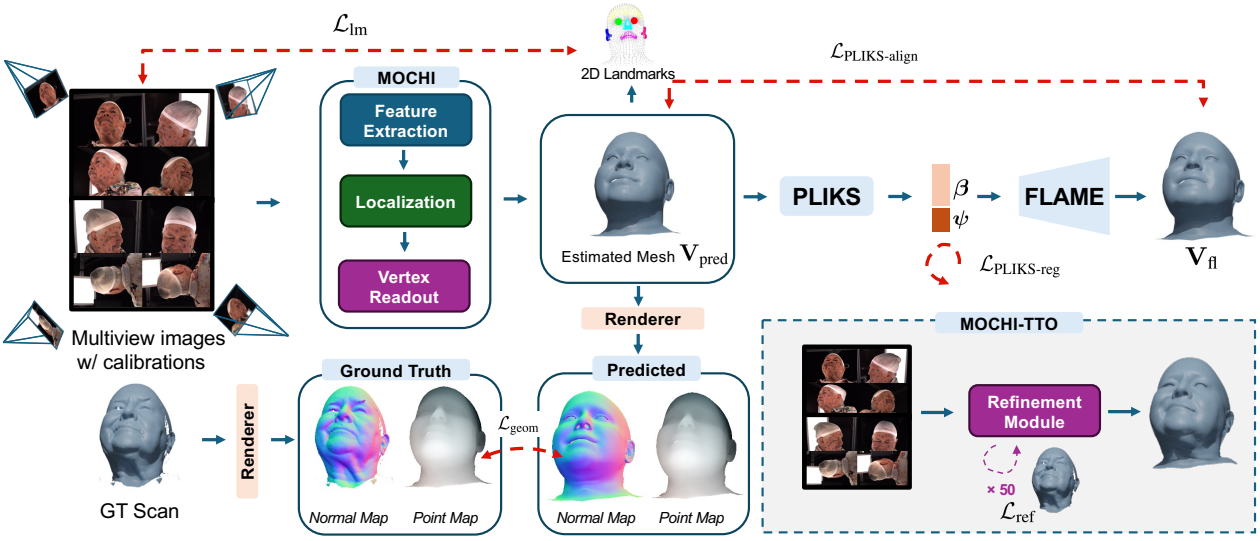

Figure 3. Given calibrated multi-view images, MOCHI predicts the vertex positions $\mathbf{V}_{\text{pred}}$ of a FLAME-topology mesh $\hat{M} = (\mathbf{V}_{\text{pred}}, \mathbf{T})$ that is spatially aligned to the raw scan. A pseudo-linear inverse kinematic solver is used to recover the FLAME identity and expression parameters that best explain $\mathbf{V}_{\text{pred}}$, which are then passed through a differentiable FLAME layer to obtain a canonical mesh with vertices $\mathbf{V}_{\text{fl}}$. We regularize the predicted mesh against $\mathbf{V}_{\text{fl}}$ via edge- and vertex-position losses, and impose additional regularization on the recovered FLAME parameters. For both $\mathbf{V}_{\text{fl}}$ and $\mathbf{V}_{\text{pred}}$, we further apply a projected 2D landmark loss using the predictions of a tracker trained purely on synthetic data. Finally, we differentiably rasterize the mesh into point maps and normal maps and supervise them with the corresponding maps rendered from the ground-truth scan. At test time, an optional TTO stage (gray box, right) fine-tunes the refinement module for a small number of iterations using the same geometric losses, further improving reconstruction accuracy.

distances often fail in regions with partial or noisy geometry (e.g., around lips, teeth, and eyelids), unless the mesh is already tightly aligned to the scan. To overcome this, we introduce a dense landmark detector to predict per-vertex 2D correspondences aligned with the FLAME topology, offering semantic supervision where geometric cues are weak.

We train the detector on 25,000 synthetic images rendered in Blender, using randomized shape, expression, pose, albedo, HDR lighting, and natural-image backgrounds. Hair is added via randomly sampled HAAR [72] styles for added realism. The detector builds on a DINOv3-Large backbone with a LoRA branch and is trained via ℓ_2 regression to predict dense 2D landmarks $\mathbf{U} \in \mathbb{R}^{n_v \times 2}$ from an input image $\mathbf{I}$. Generalization is validated on held-out synthetic data and real FaMoS scans; see Sup. Mat.

3.4. Coarse Stage

The coarse stage follows the volumetric fusion design of **TEMPEH** [9], but integrates our registration-free objectives. It predicts an initial canonical mesh from calibrated multi-view images by aggregating features into a localized 3D volume and reading out vertex positions probabilistically. We denote the full model as:

$$f_{\text{coarse}} : \{\mathbf{I}_i\}_{i=1}^{K} \longmapsto \mathbf{V}_c, \tag{1}$$

where $\mathbf{V}_c$ are the vertices of the predicted mesh M_c in canonical FLAME topology. The model comprises the following steps:

Feature Extraction and Fusion. A shared encoder–decoder extracts per-view features $\mathbf{F}_i$. These are unprojected into a 3D grid $\mathbf{G}_c$ and fused via per-channel mean and variance, forming a coarse feature volume $\mathbf{Q}_c$.

Localization. A spatial transformer estimates anisotropic scale, rotation, and translation to normalize the head region, producing a localized feature volume $\mathbf{Q}'_c$.

Vertex Readout. A 3D U-Net predicts a probability volume $\mathbf{Q}_p$ with one channel per vertex. Vertex positions are obtained by spatial soft-argmax over $\mathbf{Q}_p$, yielding the vertices $\mathbf{V}_c$ of the coarse mesh M_c.

Rendered Geometric Losses. Conventional scan-to-mesh losses rely on nearest-neighbor search (e.g., point-to-surface), which introduce *non-smooth gradients* due to discrete closest-point assignments. This often destabilizes training and may cause mesh artifacts or self-intersections, especially under large deformations. Instead, we supervise geometry using *rendered geometric maps* from all calibrated views, which provide smooth and spatially consistent gradients. Given the ground-truth scan $\mathbf{S}$, we render for each camera view i: (1) the surface normals $\mathbf{N}_{\text{gt},i}$ and (2) the per-pixel 3D point maps $\mathbf{P}_{\text{gt},i}$. From a predicted mesh $(\mathbf{V}_{\text{pred}}, \mathbf{T})$, we differentiably render the corresponding maps $\mathbf{N}_{\text{pred},i}$ and $\mathbf{P}_{\text{pred},i}$. We then minimize a robust geometric discrepancy along all views, using the Geman–McClure penalty [29] to mitigate scan noise and out-

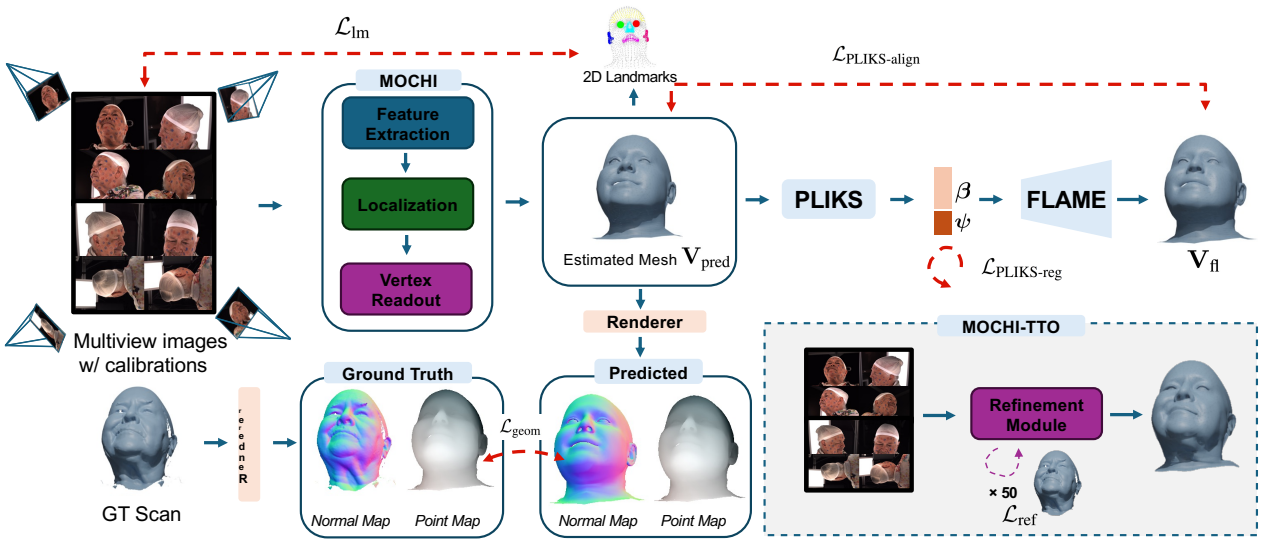

图3：给定已标定的多视角图像，MOCHI预测与原始扫描在空间上对齐的FLAME拓扑网格$\hat{M} = (\mathbf{V}_{\text{pred}}, \mathbf{T})$的顶点位置$\mathbf{V}_{\text{pred}}$。使用伪线性逆运动学求解器恢复最能解释$\mathbf{V}_{\text{pred}}$的FLAME身份和表情参数，随后通过可微FLAME层获得具有顶点$\mathbf{V}_{\text{fl}}$的规范网格。我们通过边和顶点位置损失对预测网格相对于$\mathbf{V}_{\text{fl}}$进行正则化，并对恢复的FLAME参数施加额外正则化。对于$\mathbf{V}_{\text{fl}}$和$\mathbf{V}_{\text{pred}}$，我们进一步使用仅在合成数据上训练得到的跟踪器的预测施加投影2D地标损失。最后，我们将网格可微光栅化为点图和法线图，并用从真实扫描渲染的对应图进行监督。在测试时，可选的TTO阶段（灰色框，右侧）使用相同的几何损失对细化模块进行少量迭代的微调，进一步提高重建精度。

在几何信息部分或含噪的区域（例如嘴唇、牙齿和眼睑周围），距离度量往往失效，除非网格已与扫描紧密对齐。为克服这一问题，我们引入了一个密集地标检测器，用于预测与FLAME拓扑对齐的逐顶点二维对应关系，从而在几何线索较弱的区域提供语义监督。

我们在Blender中渲染的25,000张合成图像上训练检测器，使用随机化的形状、表情、姿态、反照率、HDR光照以及自然图像背景。通过随机采样的HAAR [72]风格添加头发以增强真实感。该检测器基于DINOv3-Large骨干网络，并带有LoRA分支，通过ℓ_2回归训练，从输入图像$\mathbf{U} \in \mathbb{R}^{n_v \times 2}$预测密集的2D地标$\mathbf{I}$。泛化性能在留出的合成数据和真实FaMoS扫描上得到验证；详见补充材料。

3.4. 粗粒度阶段

粗阶段遵循TEMPEH [9]的体素融合设计，但整合了我们无配准的目标函数。它通过将特征聚合到局部3D体积中，并概率性地读出顶点位置，从已标定的多视图图像预测初始规范网格。我们将完整模型表示为：

$$f_{\text{coarse}} : \{\mathbf{I}_i\}_{i=1}^{K} \longmapsto \mathbf{V}_c, \tag{1}$$

其中 $\mathbf{V}_c$ 是规范FLAME拓扑中预测网格 M_c 的顶点。该模型包含以下部分：

以下步骤：

Feature Extraction and Fusion. 共享编码器-解码器提取每视图特征$\mathbf{F}_i$。这些特征被反投影到3D网格$\mathbf{G}_c$中，并通过逐通道均值和方差进行融合，形成粗略特征体$\mathbf{Q}_c$。

Localization. 空间Transformer估计各向异性尺度、旋转和平移，以归一化头部区域，生成局部特征体 $\mathbf{Q}'_c$。

Vertex Readout. 3D U-Net预测一个概率体$\mathbf{Q}_p$，每个顶点对应一个通道。通过对$\mathbf{Q}_p$进行空间soft-argmax得到顶点位置，从而生成粗网格M_c的顶点V_c。

渲染几何损失。传统的扫描到网格损失依赖于最近邻搜索（例如，点到表面），这会因离散最近点分配而引入*non-smooth gradients*。这常常使训练不稳定，并可能导致网格伪影或自相交，尤其是在大变形情况下。相反，我们使用来自所有标定视图的*rendered geometric maps*来监督几何，这些视图提供平滑且空间一致的梯度。给定地面真值扫描$\mathbf{S}$，我们为每个相机视图i渲染：（1）表面法线$\mathbf{N}_{\text{gt},i}$和（2）逐像素3D点图$\mathbf{P}_{\text{gt},i}$。从预测网格（$\mathbf{V}_{\text{pred}}, \mathbf{T}$）中，我们可微地渲染对应的图$\mathbf{N}_{\text{pred},i}$和$\mathbf{P}_{\text{pred},i}$。然后，我们使用Geman–McClure惩罚[29]沿所有视图最小化鲁棒几何差异，以减轻扫描噪声和异常值。

liers:

$$\mathcal{L}_{\text{geom}} = \sum_{i=1}^{K} \rho_{\text{GM}}\big(\|\mathbf{N}_{\text{pred,i}} - \mathbf{N}_{\text{gt,i}}\|_2\big) + \rho_{\text{GM}}\big(\|\mathbf{P}_{\text{pred,i}} - \mathbf{P}_{\text{gt,i}}\|_2\big), \tag{2}$$

where $\rho_{\text{GM}}(x) = \frac{x^2}{x^2+\sigma^2}$ (σ is set to 10).

Recovery of 3DMM Parameters. To anchor predicted meshes to a canonical topology, we apply a differentiable inverse module based on the PLIKS layer [70]. Given a predicted mesh $\mathbf{V}_{\text{pred}} \in \mathbb{R}^{n_v \times 3}$, the solver first estimates a set of rigid transformations $\{\mathbf{R}_s\}_{s=1}^{S}$ for FLAME's S skinning segments via Procrustes alignment between template vertices and their predicted locations (using dominant segment assignments). For efficiency, we perform this estimation only once, without iterative refinement. With rotations fixed, PLIKS solves a least-squares problem to estimate shape $\boldsymbol{\beta}$, expression $\boldsymbol{\psi}$, and translation $\mathbf{t}$:

$$[\boldsymbol{\beta}, \boldsymbol{\psi}, \mathbf{t}] = \arg\min \ \big\|\mathbf{R}\big(\bar{\mathbf{V}} + \mathbf{B}_{\text{id}}\boldsymbol{\beta} + \mathbf{B}_{\text{exp}}\boldsymbol{\psi}\big) + \mathbf{t} - \mathbf{V}_{\text{pred}}\big\|_2^2.$$

Here, $\mathbf{R}$ denotes the piecewise rigid motion applied per-vertex, using the rotation of the corresponding dominant segment. This compact notation is a simplification for clarity; the underlying system accounts for per-vertex transformations and remains fully linear in the unknowns. We then regularize the recovered parameters to encourage in-distribution identity and expression:

$$\mathcal{L}_{\text{PLIKS-reg}} = \lambda_\beta \|\boldsymbol{\beta}\|_2^2 + \lambda_\psi \|\boldsymbol{\psi}\|_2^2.$$

Re-forward and Topology Enforcement. The recovered parameters are re-forwarded through a FLAME layer [48] to obtain a topology-respecting set of vertices $\mathbf{V}_{\text{fl}}$, correcting distortions from the unconstrained regression. We then enforce vertex-level consistency and edge regularization:

$$\mathcal{L}_{\text{PLIKS-align}} = \lambda_v \big\|\mathbf{V}_{\text{fl}} - \mathbf{V}_{\text{pred}}\big\|_2^2 \ + \ \lambda_e \, \mathcal{L}_{\text{edge}}\big(\mathbf{V}_{\text{fl}}, \mathbf{V}_{\text{pred}}\big), \tag{3}$$

where $\mathcal{L}_{\text{edge}}$ penalizes relative changes of edge lengths, given the mesh triangles $\mathbf{T}$.

With these terms, PLIKS serves as an *implicit topology regularizer*: it softly aligns unconstrained vertex predictions to valid FLAME topology, while allowing gradients to flow through both representations. This bidirectional consistency encourages $\mathbf{V}_{\text{pred}}$ to follow plausible anatomical structure, and drives $\mathbf{V}_{\text{fl}}$ to reflect view-dependent details from free-form predictions.

Note: One might ask why we do not regress FLAME parameters directly. Learning identity parameters robustly would require many more subjects than typical multi-view datasets provide, leading to overfitting. Moreover, predicting FLAME parameters would constrain reconstructions strictly to its shape space. Our implicit regularization instead nudges $\mathbf{V}_{\text{pred}}$ toward FLAME's canonical topology while still allowing deviations from the 3DMM manifold.

Landmark Loss. Topology-aware and geometric losses alone cannot resolve ambiguities caused by partial, noisy, or semantically imprecise scans. To address this, we supervise the model using dense 2D landmarks from the detector (see Section 3.3), which provide a semantically rich signal over key facial regions.

Given tracked 2D landmarks $\mathbf{U}_i \in \mathbb{R}^{n_v \times 2}$ for view i and projection function Π_i, we define a reprojection loss for a set of 3D vertices $\mathbf{V} \in \mathbb{R}^{n_v \times 3}$ as $\mathcal{D}_i(\mathbf{V}) = \|(\Pi_i(\mathbf{V}) - \mathbf{U}_i)\|_2^2$. We apply this loss to both the canonical mesh $\mathbf{V}_{\text{fl}}$ and the free-form output $\mathbf{V}_{\text{pred}}$, encouraging semantic consistency in both regressed parameters and raw vertex predictions:

$$\mathcal{L}_{\text{lm}} = \sum_{i=1}^{K} \left[\mathcal{D}_i(\mathbf{V}_{\text{fl}}) + \mathcal{D}_i(\mathbf{V}_{\text{pred}})\right].$$

Total Training Objective. The overall training loss combines geometric, landmark, and topology consistency terms:

$$\begin{aligned}\mathcal{L}_{\text{total}} \ = \ & \lambda_{\text{geom}}\mathcal{L}_{\text{geom}} \ + \ \lambda_{\text{lm}}\,\mathcal{L}_{\text{lm}} \\ & + \ \lambda_{\text{align}}\,\mathcal{L}_{\text{PLIKS-align}} \ + \ \lambda_{\text{reg}}\,\mathcal{L}_{\text{PLIKS-reg}},\end{aligned} \tag{4}$$

where each λ term controls the strength of its term.

3.5. Refinement Stage

Refinement Stage. To enhance geometric detail, the refinement stage adjusts each coarse vertex based on localized multi-view features and surface cues. We denote the full refinement module as

$$f_{\text{refine}} : (\mathbf{V}_{\text{pred}}, \{\mathbf{F}_i\}_{i=1}^{K}) \mapsto \mathbf{V}_{\text{ref}},$$

where $\mathbf{V}_{pred}$ is the predicted coarse mesh and $\mathbf{F}_i$ are the already extracted per-view image features.

Following TEMPEH [9], we perform these steps:

Surface-aware fusion. For each vertex $\mathbf{v}_p \in \mathbf{V}_{pred}$, multi-view features are sampled over a local 3D grid aligned with its surface normal and aggregated using visibility-aware weights.

Probabilistic refinement. A local 3D U-Net predicts a spatial distribution over the sampling grid, and the refined position $\mathbf{v}_r$ is then obtained via soft-argmax.

To train the refinement module, now that we have a coarse registration $\mathbf{V}_{\text{ref}}$ we can use it as a geometric anchor. The total loss $\mathcal{L}_{\text{ref}}$ combines: (1) a multi-view landmark loss $\sum_{i=1}^{K} \mathcal{D}_i(\mathbf{V}_{\text{ref}})$; (2) edge-length regularization $\mathcal{L}_{\text{edge}}(\mathbf{V}_{\text{ref}}, \mathbf{V}_{\text{pred}})$; (3) an eyeball constraint $\|\mathbf{M}^{\text{eyes}} \odot (\mathbf{V}_{\text{ref}} - \mathbf{V}_{\text{pred}})\|_2^2$, where $\mathbf{M}^{\text{eyes}}$ is an eyeball-vertex mask; and (4) a geometric loss as defined in Eq. 2.

3.6. Test-Time Optimization

While MOCHI generalizes well to unseen subjects and expressions, we can further tailor a reconstruction to a *specific* capture at inference via a short *test-time optimization*

离群点：

$$\mathcal{L}_{\text{geom}} = \sum_{i=1}^{K} \rho_{\text{GM}}\big(\|\mathbf{N}_{\text{pred,i}} - \mathbf{N}_{\text{gt,i}}\|_2\big) + \rho_{\text{GM}}\big(\|\mathbf{P}_{\text{pred,i}} - \mathbf{P}_{\text{gt,i}}\|_2\big), \tag{2}$$

其中$\rho_{\text{GM}}(x) = \frac{x^2}{x^2+\sigma^2}$ (σ设为10)。

3DMM参数恢复。为了将预测的网格锚定到规范拓扑，我们应用了一个基于PLIKS层[70]的可微逆模块。给定预测的网格$\mathbf{V}_{\text{pred}} \in \mathbb{R}^{n_v \times 3}$，求解器首先通过模板顶点与其预测位置之间的Procrustes对齐（使用主导段分配），为FLAME的S蒙皮段估计一组刚性变换$\{\mathbf{R}_s\}_{s=1}^{S}$。为了提高效率，我们仅执行一次此估计，不进行迭代细化。在旋转固定后，PLIKS求解一个最小二乘问题来估计形状$\boldsymbol{\beta}$、表情$\boldsymbol{\psi}$和平移$\mathbf{t}$：

$$[\boldsymbol{\beta}, \boldsymbol{\psi}, \mathbf{t}] = \arg\min \ \big\|\mathbf{R}\big(\bar{\mathbf{V}} + \mathbf{B}_{\text{id}}\boldsymbol{\beta} + \mathbf{B}_{\text{exp}}\boldsymbol{\psi}\big) + \mathbf{t} - \mathbf{V}_{\text{pred}}\big\|_2^2.$$

此处，$\mathbf{R}$表示逐顶点应用的分段刚性运动，使用相应主导分段的旋转。这一紧凑记法是为清晰起见所作的简化；底层系统考虑逐顶点变换，并且在未知量中保持完全线性。随后，我们对恢复的参数进行正则化，以鼓励分布内的身份和表情：

$$\mathcal{L}_{\text{PLIKS-reg}} = \lambda_\beta \|\boldsymbol{\beta}\|_2^2 + \lambda_\psi \|\boldsymbol{\psi}\|_2^2.$$

重新前向与拓扑约束。将恢复的参数重新前向通过一个FLAME层[48]，以获得一组符合拓扑结构的顶点$\mathbf{V}_{\text{fl}}$，从而修正无约束回归带来的形变。随后我们施加顶点级一致性与边正则化：

$$\mathcal{L}_{\text{PLIKS-align}} = \lambda_v \big\|\mathbf{V}_{\text{fl}} - \mathbf{V}_{\text{pred}}\big\|_2^2 \ + \ \lambda_e \, \mathcal{L}_{\text{edge}}\big(\mathbf{V}_{\text{fl}}, \mathbf{V}_{\text{pred}}\big), \tag{3}$$

其中 $\mathcal{L}_{\text{edge}}$ 惩罚边长的相对变化，给定网格三角形 T。

基于这些术语，PLIKS 充当一个 *implicit topology regularizer*：它将无约束的顶点预测软对齐到有效的 FLAME 拓扑，同时允许梯度流经两种表示。这种双向一致性促使 $\mathbf{V}_{\text{pred}}$ 遵循合理的解剖结构，并驱动 $\mathbf{V}_{\text{fl}}$ 反映来自自由形式预测的视图相关细节。

Note: 有人可能会问，为何我们不直接回归FLAME参数。稳健地学习身份参数所需的受试者数量远超典型多视角数据集所能提供，从而导致过拟合。此外，预测FLAME参数会将重建严格限制在其形状空间内。我们的隐式正则化则引导$\mathbf{V}_{\text{pred}}$趋向FLAME的规范拓扑，同时仍允许偏离3DMM流形。

地标损失。仅靠拓扑感知损失和几何损失无法解决由部分、噪声或语义不精确的扫描引起的歧义。为此，我们使用检测器提供的密集2D地标（见第3.3节）来监督模型，这些地标在关键面部区域提供了语义丰富的信号。

给定视图i的已跟踪2D地标$\mathbf{U}_i \in \mathbb{R}^{n_v \times 2}$和投影函数$\Pi_i$，我们为3D顶点集$\mathbf{V} \in \mathbb{R}^{n_v \times 3}$定义重投影损失为$\mathcal{D}_i(\mathbf{V}) = \|(\Pi_i(\mathbf{V}) - \mathbf{U}_i)\|_2^2$。我们将该损失应用于规范网格$\mathbf{V}_{\text{fl}}$和自由形式输出$\mathbf{V}_{\text{pred}}$，从而在回归参数和原始顶点预测中促进语义一致性：

$$\mathcal{L}_{\text{lm}} = \sum_{i=1}^{K} \left[\mathcal{D}_i(\mathbf{V}_{\text{fl}}) + \mathcal{D}_i(\mathbf{V}_{\text{pred}})\right].$$

总训练目标。整体训练损失结合了几何、地标和拓扑一致性项：

$$\begin{aligned}\mathcal{L}_{\text{total}} \ = \ & \lambda_{\text{geom}}\mathcal{L}_{\text{geom}} \ + \ \lambda_{\text{lm}}\,\mathcal{L}_{\text{lm}} \\ & + \ \lambda_{\text{align}}\,\mathcal{L}_{\text{PLIKS-align}} \ + \ \lambda_{\text{reg}}\,\mathcal{L}_{\text{PLIKS-reg}},\end{aligned} \tag{4}$$

其中每个λ项控制其项的强度。

3.5. 精炼阶段

细化阶段。为增强几何细节，细化阶段基于局部多视图特征和表面线索调整每个粗顶点。我们将完整细化模块表示为

$$f_{\text{refine}} : (\mathbf{V}_{\text{pred}}, \{\mathbf{F}_i\}_{i=1}^{K}) \mapsto \mathbf{V}_{\text{ref}},$$

其中$\mathbf{V}_{pred}$为预测的粗网格，$\mathbf{F}_i$为已提取的逐视图图像嵌入。

按照TEMPEH [9]，我们执行以下步骤：

Surface-aware fusion. 对于每个顶点$\mathbf{v}_p \in \mathbf{V}_{pred}$，多视图特征在与其表面法线对齐的局部3D网格上进行采样，并使用可见性感知权重进行聚合。

Probabilistic refinement. 局部3D U-Net预测采样网格上的空间分布，随后通过soft-argmax获得精炼位置$\mathbf{v}_r$。

为了训练细化模块，既然我们已有粗略配准$\mathbf{V}_{\text{ref}}$，可将其用作几何锚点。总损失$\mathcal{L}_{\text{ref}}$由以下部分组成：（1）多视角地标损失$\sum_{i=1}^{K}\mathcal{D}_i(\mathbf{V}_{\text{ref}})$；（2）边长正则化$\mathcal{L}_{\text{edge}}(\mathbf{V}_{\text{ref}}, \mathbf{V}_{\text{pred}})$；（3）眼球约束$\|\mathbf{M}^{\text{eyes}} \odot (\mathbf{V}_{\text{ref}} - \mathbf{V}_{\text{pred}})\|_2^2$，其中$\mathbf{M}^{\text{eyes}}$为眼球顶点掩码；（4）如式（2）所定义的几何损失。

3.6. 测试时优化

尽管MOCHI对未见过的对象和表情具有良好的泛化能力，我们仍可通过简短的*test-time optimization*在推理时进一步针对*spe- cific*捕捉定制重建结果。

Dataset		Method	Total			Face			Lips			Neck		
			Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓
FaMoS	Images Only	*TEMPEH*	0.36	0.63	1.41	0.32	0.49	1.02	0.47	0.74	**1.10**	0.63	1.41	**2.24**
		MOCHI	**0.26**	**0.48**	**1.38**	**0.23**	**0.36**	**0.98**	**0.37**	**0.69**	1.23	**0.46**	**1.22**	2.36
	Images + Scan	*Classic Registrations*	0.10	0.24	1.40	0.09	0.15	**0.89**	0.09	0.28	**1.12**	0.23	0.98	2.79
		MOCHI TTO	**0.07**	**0.21**	**1.31**	**0.06**	**0.13**	0.92	**0.06**	**0.20**	1.22	**0.14**	**0.53**	**1.78**
CoMA	Images Only	*TEMPEH*	0.81	1.40	**1.69**	0.69	1.13	1.32	1.34	1.90	1.98	1.26	**2.27**	**2.71**
		MOCHI	**0.53**	**1.09**	1.70	**0.45**	**0.79**	**1.10**	**0.69**	**1.21**	**1.69**	**1.02**	2.31	3.21
	Images + Scan	*Classic Registrations*	0.10	0.23	0.70	0.09	0.16	0.39	0.11	0.40	1.39	0.25	0.79	1.79
		MOCHI TTO	**0.07**	**0.17**	**0.67**	**0.06**	**0.10**	**0.31**	**0.06**	**0.23**	**1.12**	**0.15**	**0.53**	**1.44**

Table 1. **Quantitative evaluation on FaMoS and CoMA.** We report Median, Mean, and Std point-to-surface error (mm) per region and for the full head (excluding scalp). Lower is better. FaMoS results use the official test split; CoMA results are zero-shot. MOCHI outperforms TEMPEH on both datasets despite training without registrations. MOCHI TTO further surpasses the classic registration pipeline.

(TTO) procedure. Given a target scan, we fine-tune *only* the refinement module for a fixed number of iterations (typically 50) using the same losses as in Sec. 3.5. This per-scan adaptation improves high-frequency detail and corrects residual misalignments – especially around challenging regions like the lips and eyelids – *without* requiring any precomputed registrations. This test-time optimization benefits from the learned priors of the local refinement module, requiring only a few iterations to converge. Unlike vertex-level fitting from scratch, our TTO operates in a learned latent space, providing semantic regularization and robustness to scan noise or partial geometry. As a result, the refinement is stable, fast, and complements the coarse-to-fine design by tailoring predictions to each specific scan.

4. Experiments

Implementation Details. We train MOCHI on one NVIDIA A100 (80GB). We pretrain using only 2D landmarks for 150,000 iterations. Next, the coarse stage is trained for 300,000 iterations, and the refinement stage for 300,000 iterations, totaling ≈1 week.

Datasets. We train and evaluate MOCHI on the **FaMoS** multi-view head capture dataset [9]. FaMoS is a dynamic multi-view head dataset with 8 color cameras, intrinsic and extrinsic parameters, and raw scans. It includes 94 subjects, each performing 28 sequences of facial expressions and head motions; all subjects wear hair nets. Following the official split [9], we use 70 subjects for training (87,000 multi-view samples), 8 for validation (1,118 multi-view samples), and 16 for testing (8,800 multi view samples). To the best of our knowledge, FaMoS is the only publicly available multi-view dataset that provides both calibrated camera parameters and raw scans. We additionally evaluate on **CoMA** [61], which shares the same 3dMD rig type as FaMoS but is much smaller in scale (12 subjects, 12 expressions, 6 views vs. 8 in FaMoS). We apply TEMPEH, MOCHI, and MOCHI-TTO directly on CoMA *without any fine-tuning*, serving as a zero-shot generalization test.

Evaluation Metrics. We follow TEMPEH [9] and GRAPE [46] and quantify reconstruction accuracy on the FaMoS test set by computing the *point-to-surface* distance from each vertex of the ground-truth scan to its closest point on the predicted 3D head surface. We report the **median**, **mean**, and **standard deviation** of this distance, separately for the *head without scalp*, *face*, *lips*, and *neck* regions. We omit the scalp region from evaluation, since the subjects in the dataset wear head caps to conceal hair, resulting in unreliable geometry in those areas (see Sup. Mat.).

4.1. Quantitative Evaluations

Generalization on New Images. We evaluate the ability of MOCHI to generalize to new multi-view samples by comparing it against TEMPEH [9], which to the best of our knowledge is the only learning-based registration method with an open-source implementation. We present quantitative results on the FaMoS test set in Table 1. As can be seen, MOCHI outperforms TEMPEH, achieving lower reconstruction errors (approximately a 28% drop in the median error for the full head without scalp), despite having removed supervision with registrations.

Dataset Registration. We further evaluate the ability of MOCHI to register an entire dataset by comparing it to a traditional registration-based FLAME fitting pipeline that relies on optimization-based alignment with semi-manual supervision [9, 48]. In Table 1 we report results obtained by running test-time optimization (TTO) for 50 iterations on the FaMoS test set. MOCHI TTO outperforms the traditional, labor-intensive registration pipeline.

Zero-Shot Generalization on CoMA. To assess generalization beyond FaMoS, we evaluate all methods zero-shot on CoMA [61], which uses the same 3dMD rig type but with 6 views instead of 8 and no fine-tuning. As shown in Table 1, MOCHI outperforms TEMPEH by a significant margin (0.53 vs. 0.81 median error for the full head), confirming that the gains transfer across datasets. MOCHI TTO again surpasses the classic registration pipeline.

Dataset		Method	Total			Face			Lips			Neck		
			Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓
FaMoS	Images Only	*TEMPEH*	0.36	0.63	1.41	0.32	0.49	1.02	0.47	0.74	**1.10**	0.63	1.41	**2.24**
		MOCHI	**0.26**	**0.48**	**1.38**	**0.23**	**0.36**	**0.98**	**0.37**	**0.69**	1.23	**0.46**	**1.22**	2.36
	Images + Scan	*Classic Registrations*	0.10	0.24	1.40	0.09	0.15	**0.89**	0.09	0.28	**1.12**	0.23	0.98	2.79
		MOCHI TTO	**0.07**	**0.21**	**1.31**	**0.06**	**0.13**	0.92	**0.06**	**0.20**	1.22	**0.14**	**0.53**	**1.78**
CoMA	Images Only	*TEMPEH*	0.81	1.40	**1.69**	0.69	1.13	1.32	1.34	1.90	1.98	1.26	**2.27**	**2.71**
		MOCHI	**0.53**	**1.09**	1.70	**0.45**	**0.79**	**1.10**	**0.69**	**1.21**	**1.69**	**1.02**	2.31	3.21
	Images + Scan	*Classic Registrations*	0.10	0.23	0.70	0.09	0.16	0.39	0.11	0.40	1.39	0.25	0.79	1.79
		MOCHI TTO	**0.07**	**0.17**	**0.67**	**0.06**	**0.10**	**0.31**	**0.06**	**0.23**	**1.12**	**0.15**	**0.53**	**1.44**

表1. FaMoS和CoMA上的定量评估。我们报告每个区域及全头（不包括头皮）的中位数、平均值和标准差点到表面误差（mm）。数值越低越好。FaMoS结果使用官方测试集划分；CoMA结果为零样本。MOCHI在两个数据集上均优于TEMPEH，尽管训练时未使用配准。MOCHI TTO进一步超越了经典配准流程。

(TTO)流程。给定一个目标扫描，我们使用与第3.5节相同的损失函数，对细化模块进行固定迭代次数（通常为50次）的微调*only*。这种逐扫描自适应提升了高频细节并修正残余错位——尤其是在嘴唇和眼睑等具有挑战性的区域——*without*无需任何预计算的配准。这种测试时优化受益于局部细化模块的学习先验，仅需少量迭代即可收敛。与从头开始的顶点级拟合不同，我们的TTO在学习的潜在空间中运行，提供语义正则化，并对扫描噪声或部分几何形状具有鲁棒性。因此，细化过程稳定、快速，并通过针对每个特定扫描定制预测来补充从粗到细的设计。

4. 实验

实现细节。我们在单块NVIDIA A100（80GB）上训练MOCHI。我们仅使用2D地标预训练150,000次迭代。接下来，粗阶段训练300,000次迭代，细化阶段训练300,000次迭代，总计≈1周。

数据集。我们在FaMoS多视角头部采集数据集[9]上训练和评估MOCHI。FaMoS是一个动态多视角头部数据集，包含8个彩色相机、内参和外参以及原始扫描数据。该数据集包含94名受试者，每人执行28组面部表情和头部运动序列；所有受试者均佩戴发网。按照官方划分[9]，我们使用70名受试者进行训练（87,000个多视角样本），8名用于验证（1,118个多视角样本），16名用于测试（8,800个多视角样本）。据我们所知，FaMoS是唯一公开可用的、同时提供标定相机参数和原始扫描数据的多视角数据集。我们还在CoMA[61]上进行了额外评估，该数据集与FaMoS使用相同的3dMD采集设备类型，但规模小得多（12名受试者，12种表情，6个视角，而FaMoS为8个视角）。我们将TEMPEH、MOCHI和MOCHI-TTO直接应用于CoMA *without any fine-tuning*，作为零样本泛化测试。

评估指标。我们遵循TEMPEH [9]和GRAPE [46] 并通过计算 *point-to-surface* 距离（从真值扫描的每个顶点到预测的3D头部表面上最近点）来量化 FaMoS 测试集上的重建精度。我们分别报告该距离在 *head without scalp*、*face*、*lips* 和 *neck* 区域的中位数、均值和标准差。由于数据集中受试者佩戴头帽以遮盖头发，导致这些区域的几何形状不可靠，因此我们将头皮区域排除在评估之外（见补充材料）。

4.1. 定量评估

新图像上的泛化。我们通过将MOCHI与TEMPEH [9]进行比较，评估其对新的多视图样本的泛化能力。据我们所知，TEMPEH是唯一具有开源实现的基于学习的配准方法。我们在表1中展示了FaMoS测试集上的定量结果。可以看出，MOCHI优于TEMPEH，实现了更低的重建误差（无头皮全头的中位误差约下降28%），尽管移除了配准监督。

数据集注册。我们进一步评估了MOCHI注册整个数据集的能力，将其与传统的基于注册的FLAME拟合流程进行比较，该流程依赖于基于优化的对齐和半手动监督[9, 48]。在表1中，我们报告了在FaMoS测试集上运行50次迭代的测试时优化（TTO）所获得的结果。MOCHI TTO优于传统的、劳动密集型的注册流程。

在CoMA上的零样本泛化。为评估超出FaMoS的泛化能力，我们在CoMA[61]上对所有方法进行零样本评估，该数据集使用相同的3dMD采集设备类型，但视图数从8个减少为6个，且不进行微调。如表1所示，MOCHI以显著优势超越TEMPEH（全头中位误差0.53对0.81），证实了这些增益可跨数据集迁移。MOCHI TTO再次超越了经典配准流程。

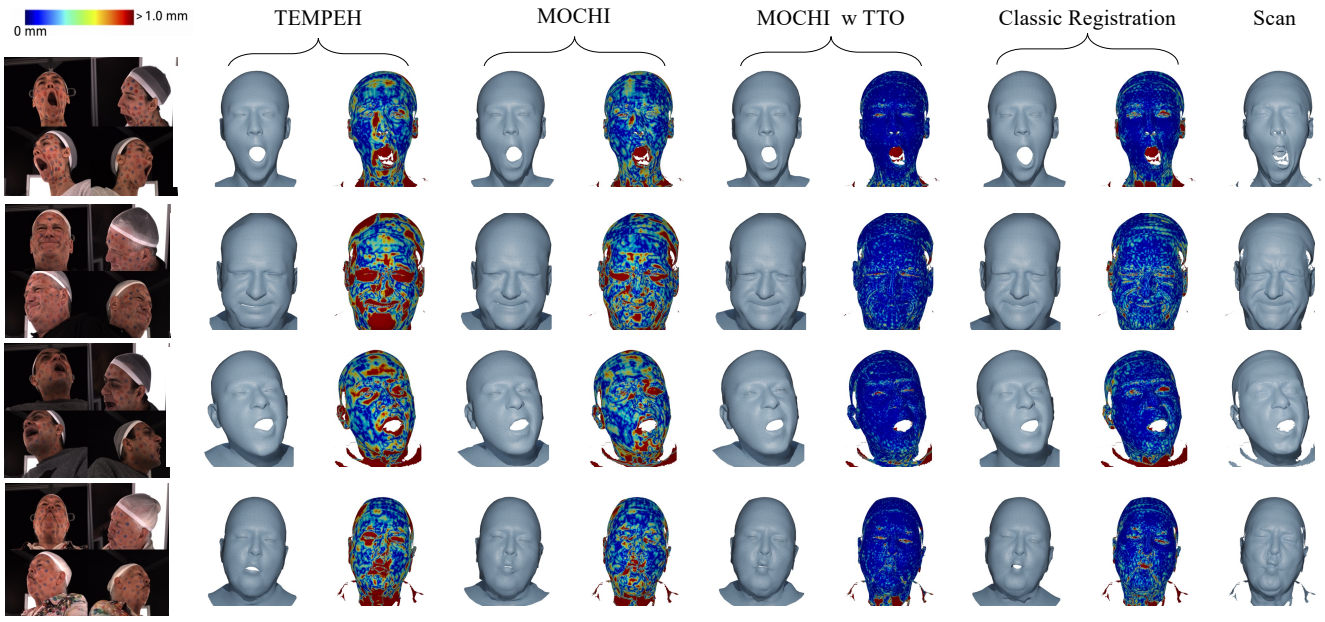

Figure 4. **Qualitative results on FaMoS (8 views).** From Left to Right: *TEMPEH* [9], *MOCHI*, *MOCHI + TTO* (50 iters), *Classic Registration*, and the *Scan*. For each method we show the predicted mesh (left) and the point-to-surface error heatmap on the scan (right; red $\geq$ 1.0 mm, lower is better). MOCHI improves over TEMPEH on unseen subjects even without using registrations for training; adding TTO further reduces error and can surpass the classic manual registration pipeline.

4.2. Qualitative Evaluation

Figure 4 presents qualitative results of both MOCHI, and MOCHI w/ TTO (50 iterations) on FaMoS test subjects, comparing against TEMPEH [9] and classic registrations. As it can be seen in the color coded point-to-surface errors, in unseen subjects MOCHI achieves a more accurate reconstruction compared to TEMPEH, even though the registration is not used for training. Additionally, MOCHI with TTO significantly improve the reconstruction result - even outperforming the classic manual registration pipeline.

4.3. Ablation Studies

Point-to-surface vs. Point maps. In Fig. 5 we compare training the coarse stage with the conventional *point-to-surface* loss against our differentiable *point-map* losses. As the point-to-surface weight increases, reconstructions develop visible artifacts and self-intersections, bypassing the topology enforcement from our regularizers. On the contrary, supervision with point-maps shows a much smoother behavior, preserving clean geometry even in larger weights.

In a similar fashion, the TTO ablations in Fig. 7 show that point maps achieve *lower* point-to-surface error than training directly with the point-to-surface objective. We believe that this is a direct result of the non-differentiable closest triangle selection in the point-to-surface loss, making the loss less suitable for optimization.

Inverse Recovery of Parameters. Fig. 6 shows the effect

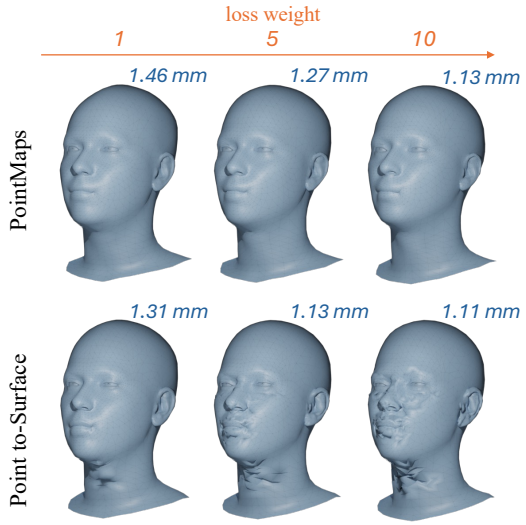

Figure 5. **Effect of loss type/weight.** Training the coarse model with point maps (top) and point-to-surface loss (bottom). When starting from a very coarse prediction using only landmark pretraining, increasing the point-to-surface loss weight leads to artifacts, whereas point-map supervision remains stable. Blue numbers show the point-to-surface error (mm).

of using only dense 2D landmarks versus using landmarks *plus* PLIKS-based inverse parameter recovery. Landmarks provide semantic guidance but do not constrain the 3D

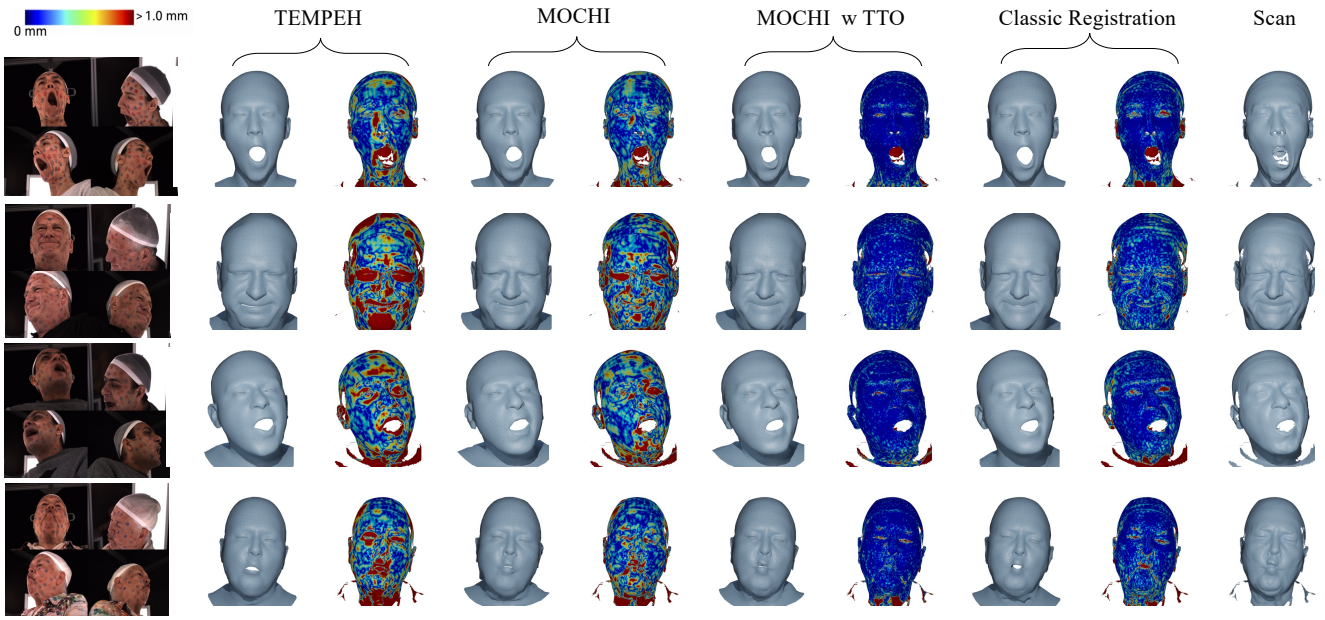

图4. FaMoS（8视角）上的定性结果。从左到右：*TEMPEH* [9]、*MOCHI*、*MOCHI + TTO* (50次迭代)、*Classic Registration*以及*Scan*。对每种方法，我们展示了预测网格（左）和扫描上的点到表面误差热图（右；红色≥ 1.0毫米，越低越好）。MOCHI在未见过的对象上优于TEMPEH，即使未使用配准进行训练；添加TTO进一步减少了误差，并可超越经典的手动配准流程。

4.2. 定性评估

图4展示了MOCHI和MOCHI w/ TTO（50次迭代）在FaMoS测试对象上的定性结果，并与TEMPEH [9]及经典配准方法进行了对比。从颜色编码的点到表面误差可以看出，在未见过的对象上，MOCHI即使未将配准用于训练，也实现了比TEMPEH更精确的重建。此外，带有TTO的MOCHI显著提升了重建效果——甚至超越了经典手动配准流程。

4.3. 消融研究

点对面与点图。在图5中，我们比较了使用传统*point-to- surface*损失与我们的可微分*point-map*损失训练粗阶段的效果。随着点对面权重的增加，重建结果出现可见的伪影和自相交，绕过了我们正则化器中的拓扑约束。相反，使用点图进行监督表现出更平滑的行为，即使在较大权重下也能保持干净的几何形状。

类似地，图7中的TTO消融实验表明，点图实现的*lower*点到表面误差低于直接使用点到表面目标训练的结果。我们认为，这直接源于点到表面损失中不可微的最邻近三角形选择，使得该损失不太适合优化。

参数的反演恢复。图6展示了其效果

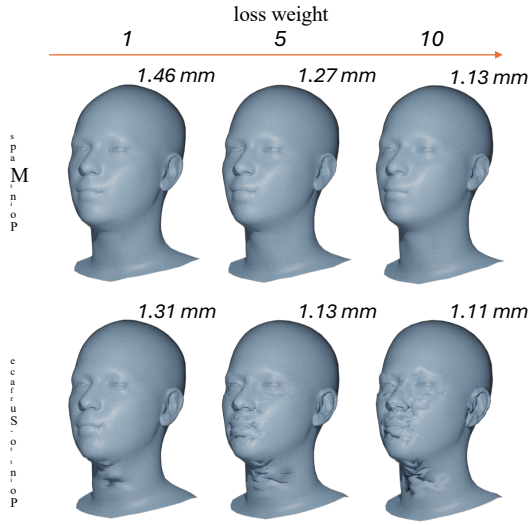

图5. 损失类型/权重的影响。使用点图（上）和点到表面损失（下）训练粗模型。当仅使用地标预训练从非常粗糙的预测开始时，增加点到表面损失权重会导致伪影，而点图监督保持稳定。蓝色数字显示点到表面误差（毫米）。

仅使用密集2D地标与使用基于PLIKS的逆参数恢复地标*plus*之间的对比。地标提供语义引导，但不对3D进行约束

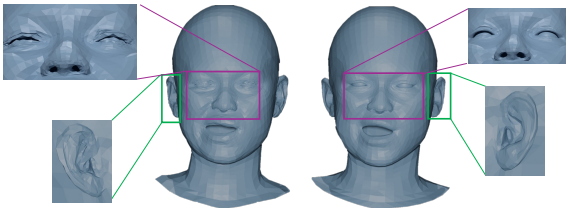

Figure 6. **Effect of inverse parameter recovery.** *Left:* only 2D landmark regularization. *Right:* 2D landmarks + PLIKS. While 2D landmarks alone can approximate the facial topology, sensitive areas such as eyes, nose, and ears exhibit artifacts and noisy triangles. Adding PLIKS enforces anatomical consistency.

topology resulting in reconstructions with self-intersecting mesh triangles and severe noise in sensitive regions (eyelids, nose/alar base, ear helix). Adding PLIKS enforces a canonical topology, suppresses artifacts, and stabilizes edges, yielding anatomically consistent eyes, nose, and lips. Note that a quantitative comparison is not meaningful since without PLIKS, landmarks alone effectively "spray" triangles onto the scan surface, yielding deceptively low errors despite severe anatomical failures. Also, omitting landmarks and supervising only with point-map losses plus PLIKS produces plausible results, but causes systematic misalignments in expression-sensitive regions (e.g., lips, eyes).

Normal Maps. Fig. 7 also presents results when performing TTO with and without normal maps. As can be seen, normal maps provide a small additional boost to reconstruction accuracy, especially in the early steps.

Direct Vertex Optimization. Instead of fine-tuning the network parameters, vertex coordinates are optimized directly to minimize the reconstruction loss. Before fine-tuning, the mesh is initialized by a single-step MOCHI prediction. For fairness, we tuned the learning rate specifically for this baseline to ensure optimal convergence. As seen in Fig. 7, direct vertex optimization also yields higher error, supporting our hypothesis that latent-space optimization provides implicit semantic regularization. Unlike direct vertex manipulation, it preserves a strong geometric prior and ensures robustness to scan noise. Note that optimizing the refinement module rather than raw vertices effectively traverses the solution manifold of plausible facial geometries, enabling fast and stable convergence with only a few dozen steps. In the Sup. Mat. we also show visually that direct vertex opt is more prone to artifacts.

Test-Time Optimization. We show results on our quantitative ablation study on test-time optimization in Fig 7. As we see, with as little as 20 iterations of TTO we are able to surpass the reconstruction accuracy of a classic manual registration pipeline. Further increasing the iterations can help improve the reconstruction result even more. Interestingly, as we already said previously, starting from the same checkpoint and training with point-to-surface for TTO results in

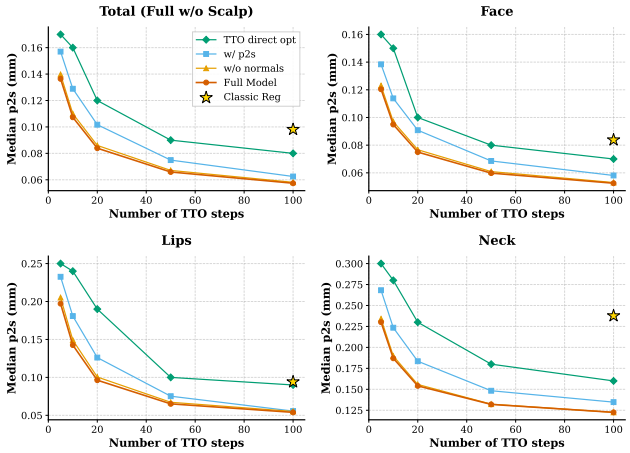

Figure 7. **TTO ablation on the FaMoS validation set.** We ablate (1) the number of TTO steps, (2) training without normals, and (3) training with the point-to-surface loss. Best results are obtained when using with both point-map and normal losses. The black star shows classic registration metrics. Full table in Sup. Mat.

suboptimal reconstruction. More extensive ablation studies and the full table can also be found in the Suppl. Mat.

Limitations. Our rendering losses outperform the point-to-surface loss but require extra computation to render point and normal maps for predictions (scan maps can be pre-rendered). On FaMoS (8 views), this adds $\approx$ 60 ms per iteration, compared to using point-to-surface loss. Additionally, the PLIKS solver adds $\approx$ 150 ms per iteration. All timings were calculated with Kaolin [40] on an NVIDIA A100 (80GB). Furthermore, despite our strong topology enforcement scheme, in rare cases TTO can fail. For instance, when scans include a large region of teeth, the model may latch onto teeth geometry and distort the lips (see Sup. Mat.).

5. Conclusions

We presented MOCHI, a registration-free framework for multi-view 3D face reconstruction in dense correspondence. By moving the registration step inside the network, MOCHI learns directly from raw multi-view scans. This is achieved by combining a pseudo-linear inverse kinematic solver to enforce a consistent FLAME topology, stable differentiable pointmap and surface-normals losses, and semantic guidance from a dense 2D landmark detector. As a result, MOCHI can be directly utilized to register new multi-view datasets from scratch using only raw scans and camera calibrations, removing the need for pre-existing manual registrations. Furthermore, a lightweight test-time optimization (TTO) can be applied to further refine the output for any given scan, leading to a final registration that is more accurate than those produced by classical, manually-intensive pipelines.

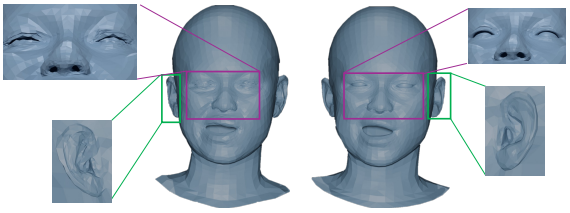

图6. 逆参数恢复的效果。*Left:*仅使用2D地标正则化。*Right:*2D地标+PLIKS。虽然仅使用2D地标可以近似面部拓扑结构，但眼睛、鼻子和耳朵等敏感区域会出现伪影和噪声三角形。添加PLIKS可强制实现解剖一致性。

拓扑结构导致重建结果中出现自相交的网格三角形，并在敏感区域（眼睑、鼻/鼻翼基底、耳轮）产生严重噪声。加入PLIKS后，强制采用规范拓扑，抑制伪影并稳定边缘，从而生成解剖学上一致的眼睛、鼻子和嘴唇。需要注意的是，定量比较并无意义，因为在不使用PLIKS的情况下，仅靠地标实际上会将三角形“喷洒”到扫描表面，尽管存在严重的解剖学失败，却产生具有欺骗性的低误差。此外，省略地标而仅使用点图损失加PLIKS进行监督，虽能产生看似合理的结果，但会在表情敏感区域（如嘴唇、眼睛）导致系统性错位。

法线贴图。图7还展示了使用和不使用法线贴图进行TTO的结果。可以看出，法线贴图对重建精度提供了小幅额外提升，尤其是在早期步骤中。

直接顶点优化。不同于微调网络参数，该方法直接优化顶点坐标以最小化重建损失。在微调之前，网格通过单步MOCHI预测进行初始化。为保证公平性，我们专门为此基线调整了学习率，以确保最优收敛。如图7所示，直接顶点优化同样产生更高的误差，这支持了我们的假设：潜在空间优化提供了隐式语义正则化。与直接顶点操作不同，它保留了强几何先验，并确保对扫描噪声的鲁棒性。值得注意的是，优化精化模块而非原始顶点，实际上是在遍历合理面部几何的解流形，从而仅需几十步即可实现快速且稳定的收敛。在补充材料中，我们还直观地展示了直接顶点优化更容易产生伪影。

测试时优化。我们在图7中展示了关于测试时优化的定量消融实验结果。可以看出，仅需20次TTO迭代，我们就能超越经典手动配准流程的重建精度。进一步增加迭代次数还能继续提升重建结果。有趣的是，正如我们之前所述，从相同的检查点出发，并使用点到面训练进行TTO，会导致

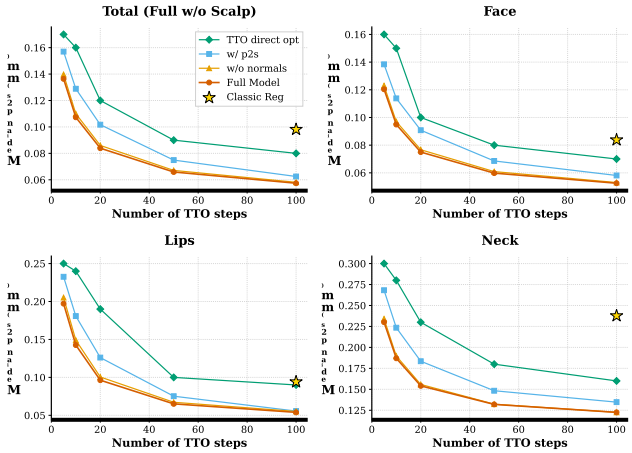

图7：在FaMoS验证集上的TTO消融实验。我们消融了(1)TTO步数、(2)不使用法向量的训练，以及(3)使用点到面损失的训练。当同时使用点图和法向量损失时获得最佳结果。黑色星号表示经典配准指标。完整表格见补充材料。

次优重建。更广泛的消融研究和完整表格也可在补充材料中找到。

局限性。我们的渲染损失优于点到表面损失，但需要额外计算来渲染预测的点图和法线图（扫描图可预先渲染）。在FaMoS（8视角）上，这比使用点到表面损失每次迭代增加≈ 60毫秒。此外，PLIKS求解器每次迭代增加≈ 150毫秒。所有计时均使用Kaolin [40]在NVIDIA A100（80GB）上计算。此外，尽管我们采用了强拓扑约束方案，但在少数情况下TTO仍可能失败。例如，当扫描包含大面积牙齿区域时，模型可能锁定牙齿几何形状并扭曲嘴唇（见补充材料）。

5. 结论

我们提出了MOCHI，一种用于密集对应多视角三维人脸重建的无配准框架。通过将配准步骤移入网络内部，MOCHI直接从原始多视角扫描中学习。这通过结合伪线性逆运动学求解器以强制一致的FLAME拓扑、稳定的可微点图和表面法线损失，以及来自密集二维地标检测器的语义引导来实现。因此，MOCHI可直接用于从零开始配准新的多视角数据集，仅需原始扫描和相机标定，无需预先存在的手动配准。此外，轻量级的测试时优化（TTO）可应用于任何给定扫描以进一步细化输出，最终得到的配准结果比经典的手动密集型流程所产生的结果更为精确。

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Supplementary Material

This supplementary material provides additional details and results for MOCHI.

A. Implementation Details

FaMoS Setup The FaMoS dataset acquisition setup comprises both grayscale stereo pairs (8 pairs) and RGB cameras (8 cameras). For training MOCHI, we rely exclusively on the 8 RGB cameras; an example of the input views from this configuration is shown in Fig. 8. To ensure a fair comparison, we also retrained the baseline method TEMPEH[9] using its official open-source implementation, configuring it to use the same set of 8 RGB cameras instead of the stereo pairs. For more information on FaMoS please see [9].

Figure 8. **Multi-view input setup.** Example views from the 8 RGB cameras used in the FaMoS dataset. We utilize only these RGB streams for both MOCHI and the TEMPEH baseline.

Exclusion of Scalp Region from Evaluation In the quantitative evaluation presented in the main text, we explicitly exclude the scalp region from our evaluation protocol. This exclusion is necessary because subjects in the FaMoS dataset wore hair caps (or hair nets) to conceal their hair during capture. These caps introduce noise into the reconstructed raw scans—often capturing the geometry of the fabric rather than the subject's actual anatomical shape. Consequently, the raw scans in this region do not serve as a reliable ground truth for modeling the scalp. We illustrate examples of this in Fig. 9.

Evaluation Regions. We adopt the official vertex masks provided with the FLAME model [48] to define our regions of interest. Fig. 10 visualizes the specific vertex subsets used in our quantitative analysis. The regions highlighted in **blue** (Lips, Face, Neck, and the combined Head) are used for calculating reconstruction errors. Conversely, the regions highlighted in **red** are excluded from evaluation. Specifically, we omit the scalp region due to the capture artifacts discussed previously, and the lower mesh boundary that are not relevant to facial reconstruction quality and can be affected by non-relevant to the head scan noise.

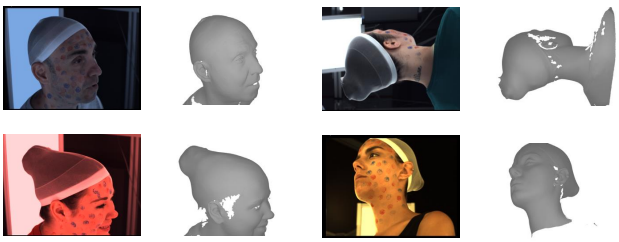

Figure 9. **Impact of hair caps on ground-truth scans.** We show example input images and their corresponding raw scans from the FaMoS dataset. Depending on the fit of the hair cap, the resulting scans exhibit significant distortion in the scalp region, often including the geometry of the cap itself. As a result, this region does not faithfully represent the subject's actual scalp and is excluded from our quantitative metrics.

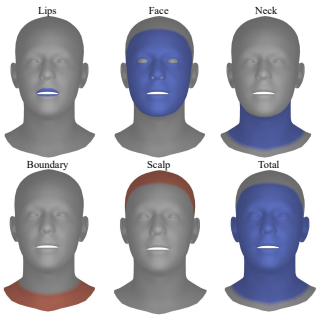

Figure 10. **Visualization of evaluation masks.** We utilize standard FLAME vertex masks to compute regional metrics. **Blue regions** indicate vertices included in the evaluation: (Top Row) Lips, Face, and Neck; (Bottom Right) the combined Head mask. **Red regions** indicate excluded vertices: (Bottom Left) the lower boundary and (Bottom Center) the scalp region, which are removed to ensure robust evaluation against capture noise and to ignore regions non-relevant to face reconstruction.

B. Extended Ablation Studies

Extra Qualitative Comparisons. Figure 11 presents additional qualitative results on FaMoS validation and test subjects, comparing MOCHI (with and without TTO at 50 iterations) against TEMPEH [9] and classic registrations. We show a broad range of examples and expressions, further demonstrating MOCHI's accuracy and generalization.

Test-Time Optimization (TTO) Strategies. In Table 2, we provide the complete set of numerical results corresponding to Fig. 7 in the main text. Furthermore, we introduce the **No-Pretrain** baseline where the local refinement module is randomly initialized and optimized from scratch for each scan, instead of being initialized from pre-trained MOCHI weights.

Without MOCHI initialization (No-Pretrain), performance degrades significantly, underlining the importance of learned priors.

补充材料

本补充材料提供了MOCHI的更多细节和结果。

A. 实现细节

FaMoS 设置 FaMoS 数据集采集设置包括灰度立体像对（8 对）和 RGB 相机（8 台）。为训练 MOCHI，我们仅依赖 8 台 RGB 相机；该配置的输入视图示例见图 8 。为确保公平比较，我们还使用基线方法 TEMPEH[9] 的官方开源实现对其进行了重新训练，将其配置为使用相同的 8 台 RGB 相机而非立体像对。有关 FaMoS 的更多信息，请参阅文献 [9]。

图8：多视角输入设置。FaMoS数据集中8个RGB摄像头的示例视图。我们仅使用这些RGB流用于MOCHI和TEMPEH基线。

在定量评估中，我们将头皮区域明确排除在评估方案之外。这一排除是必要的，因为FaMoS数据集中的受试者在采集过程中佩戴了发帽（或发网）以遮盖头发。这些发帽会在重建的原始扫描中引入噪声——通常捕捉到的是织物的几何形状，而非受试者真实的解剖形态。因此，该区域的原始扫描无法作为头皮建模的可靠真值。我们在图9中展示了相关示例。

评估区域。我们采用FLAME模型[48]提供的官方顶点掩码来定义感兴趣区域。图10可视化了定量分析中使用的特定顶点子集。蓝色高亮区域（嘴唇、面部、颈部及组合头部）用于计算重建误差。相反，红色高亮区域被排除在评估之外。具体而言，由于先前讨论的采集伪影，我们省略了头皮区域，以及与人脸重建质量无关且可能受头部扫描噪声影响的下网格边界。

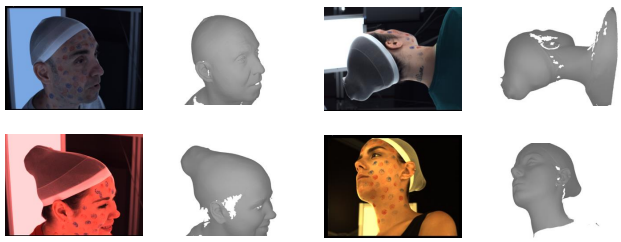

图9：发帽对真实扫描的影响。我们展示了FaMoS数据集中的示例输入图像及其对应的原始扫描。根据发帽的贴合程度，生成的扫描在头皮区域会出现显著变形，通常还包括发帽本身的几何形状。因此，该区域不能真实反映受试者的实际头皮，在我们的定量指标中被排除在外。

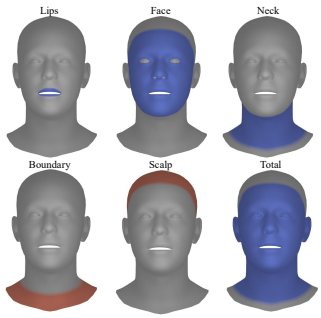

图10：评估掩码的可视化。我们使用标准FLAME顶点掩码来计算区域指标。蓝色区域表示评估中包含的顶点：（顶行）嘴唇、面部和颈部；（右下）组合头部掩码。红色区域表示排除的顶点：（左下）下边界和（下中）头皮区域，这些区域被移除，以确保评估对采集噪声具有鲁棒性，并忽略与面部重建无关的区域。

B. 扩展消融研究

额外的定性比较。图11展示了在FaMoS验证集和测试集受试者上的额外定性结果，将MOCHI（在50次迭代时使用和不使用TTO）与TEMPEH [9]及经典配准方法进行比较。我们展示了广泛的示例和表情，进一步证明了MOCHI的准确性和泛化能力。

测试时优化（TTO）策略。在表2中，我们提供了与正文中图7对应的完整数值结果集。此外，我们引入了无预训练基线，其中局部细化模块被随机初始化，并针对每次扫描从头开始优化，而非从预训练的MOCHI权重初始化。

在没有MOCHI初始化（无预训练）的情况下，性能显著下降，凸显了学习先验的重要性。

Name	Full w/o Scalp			Face			Lips			Neck		
	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓
w/ p2s t=5	0.16	0.31	1.61	0.14	0.25	1.52	0.23	0.47	1.23	0.27	0.64	1.31
w/ p2s t=10	0.13	0.27	1.58	0.11	0.21	1.50	0.18	0.38	1.15	0.22	0.56	1.24
w/ p2s t=20	0.10	0.23	1.57	0.09	0.18	1.50	0.13	0.28	0.99	0.18	0.49	1.16
w/ p2s t=50	0.07	0.18	1.55	0.07	0.15	1.49	0.08	0.19	0.89	0.15	0.42	1.08
w/ p2s t=100	**0.06**	**0.16**	1.54	0.06	0.14	1.49	0.06	**0.15**	**0.85**	0.13	0.39	1.04
w/o normals t=5	0.14	0.31	1.60	0.12	0.23	1.47	0.21	0.47	1.27	0.23	0.58	1.29
w/o normals t=10	0.11	0.26	1.57	0.10	0.20	1.44	0.15	0.39	1.22	0.19	0.49	1.16
w/o normals t=20	0.09	0.22	1.55	0.08	0.17	1.43	0.10	0.31	1.19	0.16	0.43	1.08
w/o normals t=50	0.07	0.19	1.52	0.06	0.15	1.41	0.07	0.23	1.13	0.13	0.39	1.04
w/o normals t=100	**0.06**	0.17	1.51	**0.05**	**0.13**	1.40	0.06	0.20	1.10	**0.12**	**0.37**	**1.03**
no-pretrain t=5	1.13	1.46	1.49	1.06	1.31	1.14	1.21	1.58	1.59	1.66	2.61	2.97
no-pretrain t=10	1.02	1.38	1.50	0.94	1.23	1.15	1.12	1.55	1.63	1.51	2.54	3.03
no-pretrain t=20	0.75	1.16	1.47	0.68	1.02	1.12	0.89	1.36	1.58	1.12	2.23	2.94
no-pretrain t=50	0.44	0.94	1.46	0.39	0.81	**1.10**	0.56	1.12	1.56	0.69	1.85	2.80
no-pretrain t=100	0.32	0.89	1.48	0.28	0.77	1.15	0.38	1.02	1.56	0.5	1.67	2.73
TTO direct opt t=5	0.17	0.36	1.57	0.16	0.27	1.41	0.25	0.60	1.37	0.30	0.97	1.91
TTO direct opt t=10	0.16	0.33	1.56	0.15	0.26	1.41	0.24	0.56	1.34	0.28	0.92	1.90
TTO direct opt t=20	0.12	0.29	1.55	0.10	0.23	1.41	0.19	0.50	1.31	0.23	0.85	1.88
TTO direct opt t=50	0.09	0.26	1.53	0.08	0.22	1.41	0.10	0.40	1.23	0.18	0.76	1.88
TTO direct opt t=100	0.08	0.28	1.54	0.07	0.26	1.46	0.09	0.37	1.14	0.16	0.82	2.17
MOCHI TTO (full model) t=5	0.14	0.30	1.60	0.12	0.23	1.49	0.20	0.46	1.27	0.23	0.57	1.26
MOCHI TTO (full model) t=10	0.11	0.26	1.58	0.09	0.20	1.46	0.14	0.37	1.22	0.19	0.48	1.14
MOCHI TTO (full model) t=20	0.08	0.22	1.56	0.08	0.17	1.45	0.10	0.30	1.19	0.15	0.42	1.08
MOCHI TTO (full model) t=50	0.07	0.18	1.53	0.06	0.14	1.43	0.07	0.23	1.13	0.13	0.38	1.04
MOCHI TTO (full model) t=100	**0.06**	**0.16**	1.52	**0.05**	**0.13**	1.42	**0.05**	0.19	1.09	**0.12**	**0.37**	**1.03**
Classic Registrations	0.10	0.24	**1.44**	0.08	0.16	1.21	0.09	0.36	1.39	0.24	0.90	2.39

Table 2. **Quantitative ablation of Test-Time Optimization (TTO).** We evaluate the impact of different loss functions, supervision signals, and optimization strategies on the FaMoS validation set. We report the **Median**, **Mean**, and **Std** of the point-to-surface error (mm) for the full head (excluding scalp) and specific facial regions. *Direct Opt* refers to initializing with MOCHI and optimizing vertices directly—rather than fine-tuning the parameters of the refinement model—while *No-Pretrain* optimizes the network from random initialization. Our full model (MOCHI TTO) outperforms all baselines and surpasses the quality of classical registrations.

Additionally, Fig. 13 shows how reconstruction quality improves progressively from step 5 to 100.

Direct Vertex Optimization Fig. 12 illustrates the robustness of our TTO scheme. Under identical conditions, direct vertex optimization leads to high-frequency artifacts, while our latent TTO produces smooth, topologically valid surfaces.

Chamfer Distance While prior methods [9, 46] primarily rely on point-to-surface error, the Chamfer distance is another prevalent metric in 3D vision. However, similar to point-to-surface error, the Chamfer distance relies on non-differentiable nearest-neighbor selection, which can cause artifacts and can make training more unstable.

In Fig. 14, we compare the training dynamics of the coarse stage over 500 iterations using three different objectives: the conventional point-to-surface loss, the Chamfer loss, and our differentiable point-map loss. As the weight of the point-to-surface loss increases, reconstructions rapidly develop visible artifacts and self-intersections, effectively bypassing the topology enforcement of our regularizers. Similarly, while the Chamfer distance avoids the severe topological collapses seen with point-to-surface loss, it still induces artifacts and noise (e.g., in the ears and eyeballs). In contrast, supervision with point-maps exhibits significantly smoother behavior, preserving clean geometry and stable gradients even at higher loss weights.

Limitations We visualize typical failure cases of our method in Fig. 15. While MOCHI TTO generally refines geometry, it can encounter difficulties when the raw scan contains teeth, which are not represented in the FLAME model. Additionally, large head poses can occasionally lead to artifacts in the base model. Such cases can happen more often when we train MOCHI without 2D supervision from the predicted landmarks of the dense landmark tracker.

C. Additional Dataset Evaluation

FaceScape. To further validate generalization, we evaluate MOCHI on FaceScape [91], which uses a different capture rig with per-sample self-calibration via structure-from-motion. This results in inconsistent camera parameters and arbitrary scale across samples. To evaluate on FaceScape, we align scans to a canonical frame and fine-tune our model. Since the official registrations are also unaligned, we rigidly

	Full w/o Scalp			Face			Lips			Neck		
Name	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓	Median ↓	Mean ↓	Std ↓
w/ p2s t=5	0.16	0.31	1.61	0.14	0.25	1.52	0.23	0.47	1.23	0.27	0.64	1.31
w/ p2s t=10	0.13	0.27	1.58	0.11	0.21	1.50	0.18	0.38	1.15	0.22	0.56	1.24
w/ p2s t=20	0.10	0.23	1.57	0.09	0.18	1.50	0.13	0.28	0.99	0.18	0.49	1.16
w/ p2s t=50	0.07	0.18	1.55	0.07	0.15	1.49	0.08	0.19	0.89	0.15	0.42	1.08
w/ p2s t=100	**0.06**	**0.16**	1.54	0.06	0.14	1.49	0.06	**0.15**	**0.85**	0.13	0.39	1.04
w/o normals t=5	0.14	0.31	1.60	0.12	0.23	1.47	0.21	0.47	1.27	0.23	0.58	1.29
w/o normals t=10	0.11	0.26	1.57	0.10	0.20	1.44	0.15	0.39	1.22	0.19	0.49	1.16
w/o normals t=20	0.09	0.22	1.55	0.08	0.17	1.43	0.10	0.31	1.19	0.16	0.43	1.08
w/o normals t=50	0.07	0.19	1.52	0.06	0.15	1.41	0.07	0.23	1.13	0.13	0.39	1.04
w/o normals t=100	**0.06**	0.17	1.51	**0.05**	**0.13**	1.40	0.06	0.20	1.10	**0.12**	**0.37**	**1.03**
no-pretrain t=5	1.13	1.46	1.49	1.06	1.31	1.14	1.21	1.58	1.59	1.66	2.61	2.97
no-pretrain t=10	1.02	1.38	1.50	0.94	1.23	1.15	1.12	1.55	1.63	1.51	2.54	3.03
no-pretrain t=20	0.75	1.16	1.47	0.68	1.02	1.12	0.89	1.36	1.58	1.12	2.23	2.94
no-pretrain t=50	0.44	0.94	1.46	0.39	0.81	**1.10**	0.56	1.12	1.56	0.69	1.85	2.80
no-pretrain t=100	0.32	0.89	1.48	0.28	0.77	1.15	0.38	1.02	1.56	0.5	1.67	2.73
TTO direct opt t=5	0.17	0.36	1.57	0.16	0.27	1.41	0.25	0.60	1.37	0.30	0.97	1.91
TTO direct opt t=10	0.16	0.33	1.56	0.15	0.26	1.41	0.24	0.56	1.34	0.28	0.92	1.90
TTO direct opt t=20	0.12	0.29	1.55	0.10	0.23	1.41	0.19	0.50	1.31	0.23	0.85	1.88
TTO direct opt t=50	0.09	0.26	1.53	0.08	0.22	1.41	0.10	0.40	1.23	0.18	0.76	1.88
TTO direct opt t=100	0.08	0.28	1.54	0.07	0.26	1.46	0.09	0.37	1.14	0.16	0.82	2.17
MOCHI TTO (full model) t=5	0.14	0.30	1.60	0.12	0.23	1.49	0.20	0.46	1.27	0.23	0.57	1.26
MOCHI TTO (full model) t=10	0.11	0.26	1.58	0.09	0.20	1.46	0.14	0.37	1.22	0.19	0.48	1.14
MOCHI TTO (full model) t=20	0.08	0.22	1.56	0.08	0.17	1.45	0.10	0.30	1.19	0.15	0.42	1.08
MOCHI TTO (full model) t=50	0.07	0.18	1.53	0.06	0.14	1.43	0.07	0.23	1.13	0.13	0.38	1.04
MOCHI TTO (full model) t=100	**0.06**	**0.16**	1.52	**0.05**	**0.13**	1.42	**0.05**	0.19	1.09	**0.12**	**0.37**	**1.03**
Classic Registrations	0.10	0.24	**1.44**	0.08	0.16	1.21	0.09	0.36	1.39	0.24	0.90	2.39

表2. 测试时优化（TTO）的定量消融实验。我们在FaMoS验证集上评估了不同损失函数、监督信号和优化策略的影响。我们报告了全头（不含头皮）及特定面部区域的点到表面误差（mm）的中位数、均值和标准差。*Direct Opt*指使用MOCHI初始化并直接优化顶点——而非微调细化模型的参数——而*No-Pretrain*则从随机初始化开始优化网络。我们的完整模型（MOCHI TTO）优于所有基线，并超越了经典配准的质量。

此外，图13展示了重建质量如何从第5步到第100步逐步提升。

图12展示了我们TTO方案的鲁棒性。在相同条件下，直接顶点优化会产生高频伪影，而我们的潜在TTO则生成平滑且拓扑有效的表面。

倒角距离 尽管先前的方法[9, 46]主要依赖于点到表面误差，倒角距离是3D视觉中另一种常用的度量标准。然而，与点到表面误差类似，倒角距离依赖于不可微的最近邻选择，这可能导致伪影并使训练更加不稳定。

在图14中，我们比较了粗阶段在500次迭代中使用三种不同目标的训练动态：传统的点到表面损失、Chamfer损失以及我们的可微点图损失。随着点到表面损失权重的增加，重建结果迅速出现可见的伪影和自交，有效绕过了我们正则化器的拓扑约束。同样，虽然Chamfer距离避免了严重的

与点到面损失中观察到的拓扑塌陷相比，它仍然会引入伪影和噪声（例如在耳朵和眼球区域）。相比之下，使用点图进行监督表现出显著更平滑的行为，即使在更高的损失权重下也能保持干净的几何形状和稳定的梯度。

局限性 我们在图15中可视化了我们方法的典型失败案例。尽管MOCHI TTO通常能细化几何形状，但当原始扫描包含牙齿时，它可能会遇到困难，因为牙齿未在FLAME模型中表示。此外，较大的头部姿态偶尔会导致基础模型中出现伪影。当我们不使用密集地标跟踪器预测的地标所提供的2D监督来训练MOCHI时，此类情况可能更频繁地发生。

C. 附加数据集评估

FaceScape。为进一步验证泛化能力，我们在FaceScape[91]上评估MOCHI，该数据集使用不同的采集装置，并通过运动恢复结构进行逐样本自标定。这导致不同样本间的相机参数不一致且尺度任意。为在FaceScape上进行评估，我们将扫描数据对齐到规范坐标系，并对模型进行微调。由于官方配准结果也未对齐，我们刚性

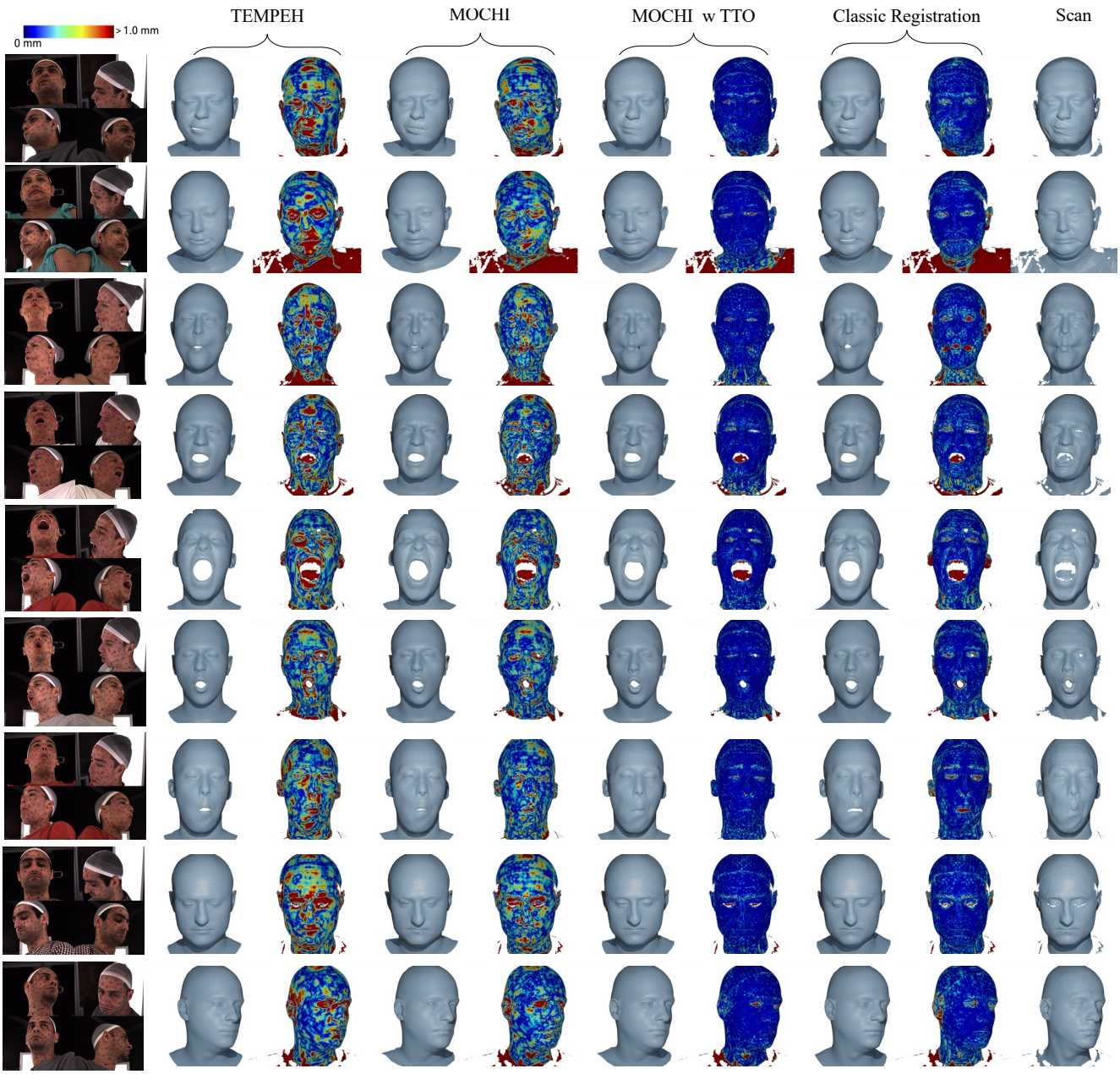

Figure 11. **Qualitative results on FaMoS validation and test sets (we show 4 of 8 input views).** From Left to Right: *TEMPEH* [9], *MOCHI*, *MOCHI + TTO* (50 iters), *Classic Registration*, and the *Scan*. For each method we show the predicted mesh (left) and the point-to-surface error heatmap on the scan (right; red $\geq$ 1.0 mm, lower is better). MOCHI improves over TEMPEH on unseen subjects even without using registrations for training; adding TTO further reduces error and can surpass the classic manual registration pipeline.

align them to the scans for fair comparison. Qualitative results comparing MOCHI, MOCHI-TTO, and the official FaceScape registration are shown in Fig. 16. As can be seen, MOCHI-TTO produces registrations that are more accurate than the official FaceScape registrations, confirming that our method generalizes beyond the FaMoS capture setup.

D. Dense Landmark Tracker

D.1. Synthetic Data With Blender

Motivation. As described in the Method section of the main paper, our approach relies on an accurate dense landmark tracker for supervision. There are several ways to obtain high-quality dense landmark datasets:

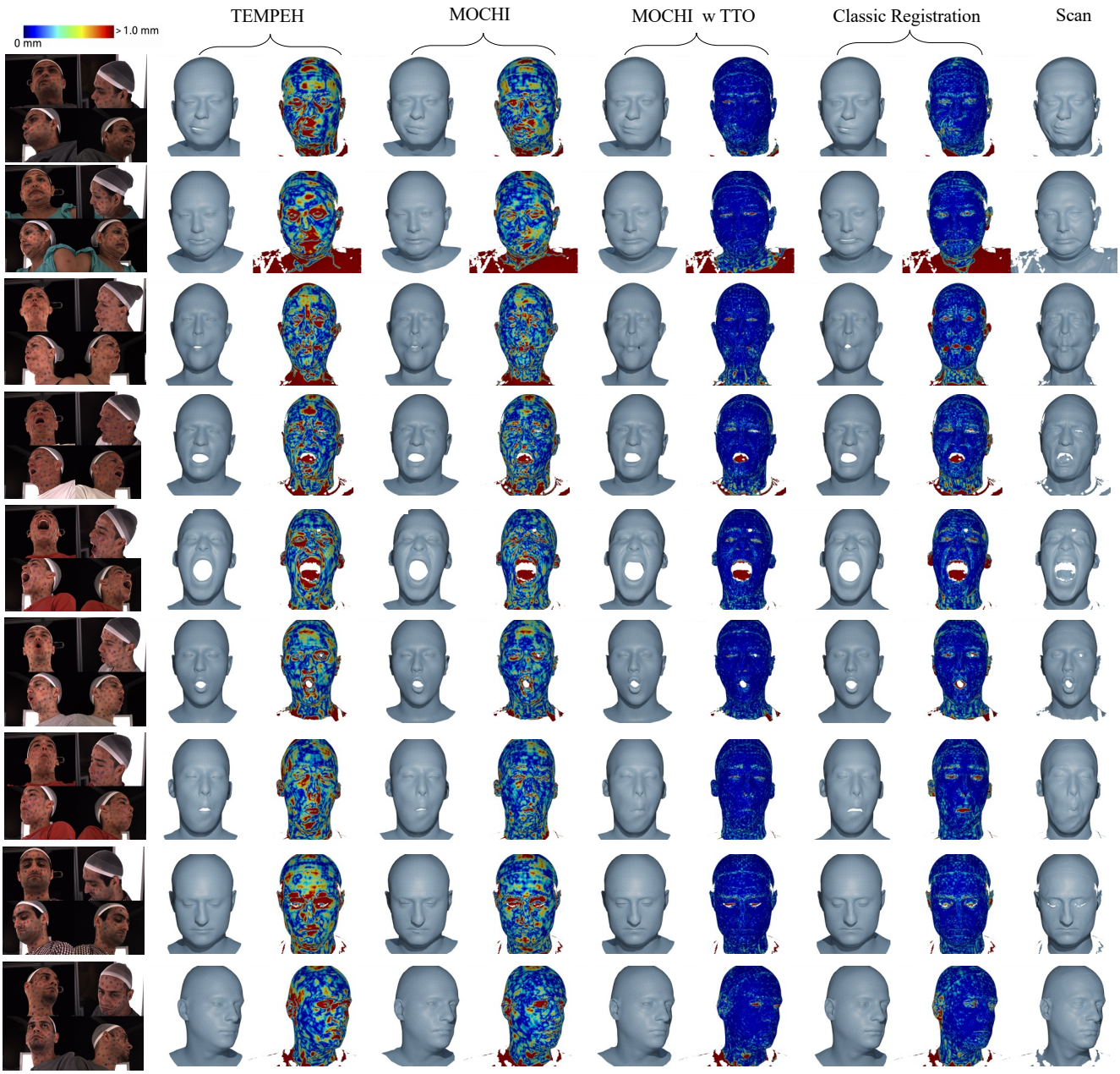

图11. FaMoS验证集和测试集上的定性结果（我们展示8个输入视角中的4个）。从左到右：*TEMPEH* [9]、*MOCHI*、*MOCHI + TTO* (50次迭代)、*Classic Registration*和*Scan*。对于每种方法，我们展示预测网格（左）和扫描上的点到表面误差热图（右；红色≥ 1.0毫米，越低越好）。MOCHI在未见过的受试者上优于TEMPEH，即使不使用配准进行训练；添加TTO进一步减少误差，并能超越经典的手动配准流程。

将它们与扫描结果对齐，以便进行公平比较。图16展示了MOCHI、MOCHI-TTO与官方FaceScape配准结果的定性对比。可以看出，MOCHI-TTO生成的配准结果比官方FaceScape配准更为精确，证实了我们的方法能够泛化到FaMoS采集设置之外。

D. 密集地标跟踪器

D.1. 使用Blender生成合成数据

动机。如主论文方法部分所述，我们的方法依赖于精确的密集地标跟踪器进行监督。有几种方法可以获得高质量的密集地标数据集：

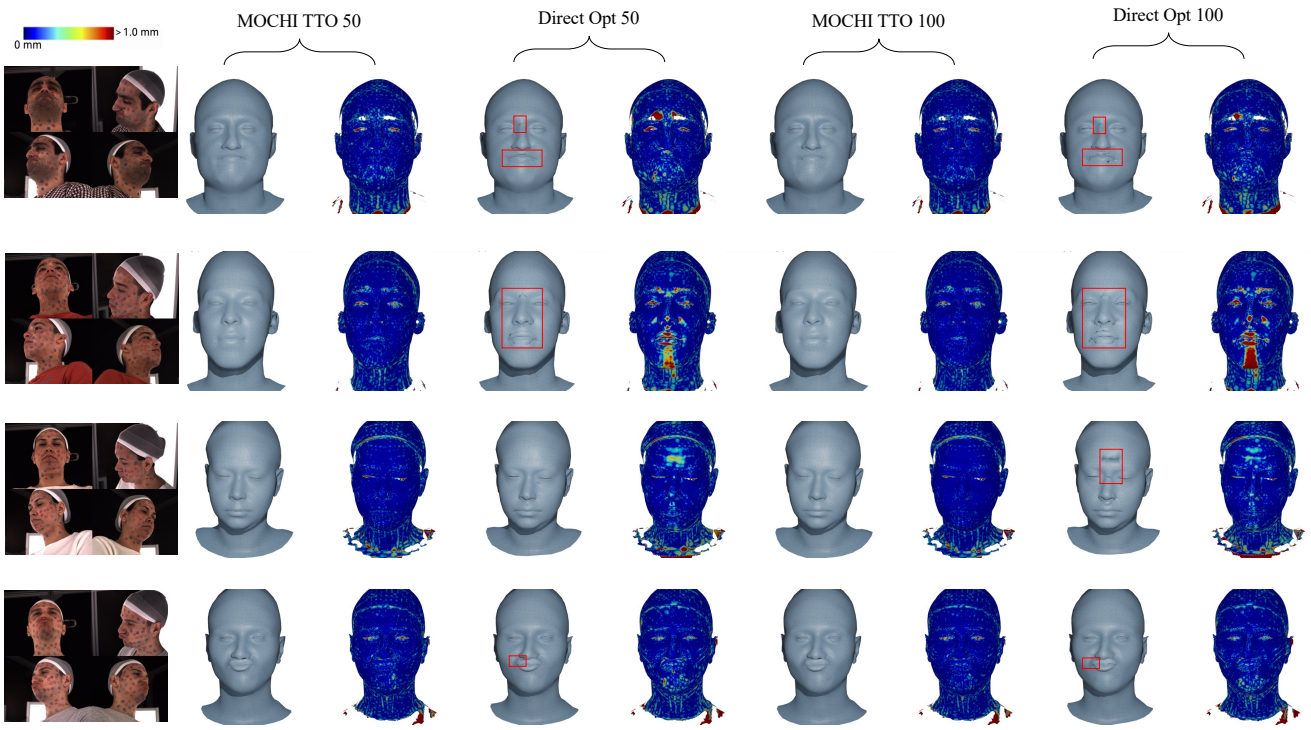

Figure 12. **Robustness of optimization strategies.** We compare our latent-space optimization (MOCHI TTO) against directly optimizing mesh vertices (Direct Opt) at 50 and 100 iterations. While Direct Opt reduces the numerical point-to-surface error, it is prone to artifacts, spikes, and surface irregularities (highlighted by red boxes). In contrast, MOCHI TTO leverages the network's learned prior, ensuring the reconstruction remains smooth and anatomically plausible while accurately fitting the target scan.

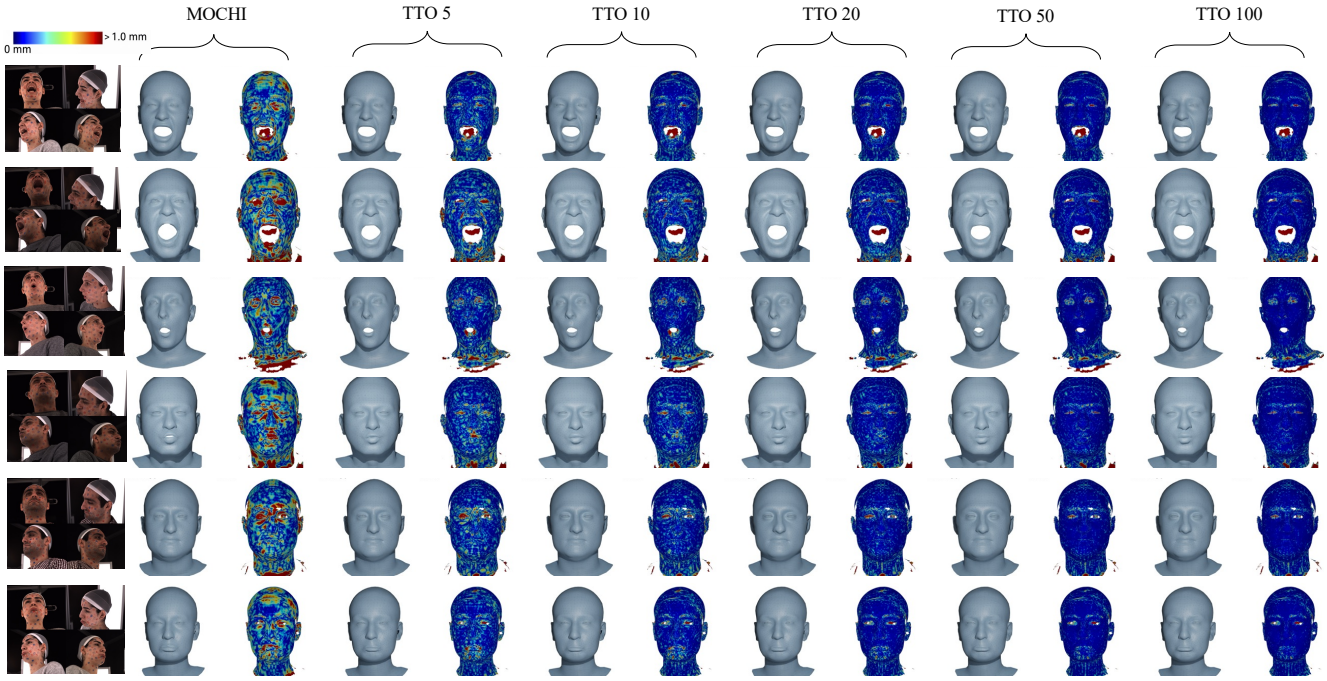

Figure 13. **Qualitative progression of Test-Time Optimization.** This figure shows the evolution of mesh refinement over 100 iterations on example images from FaMoS test set.

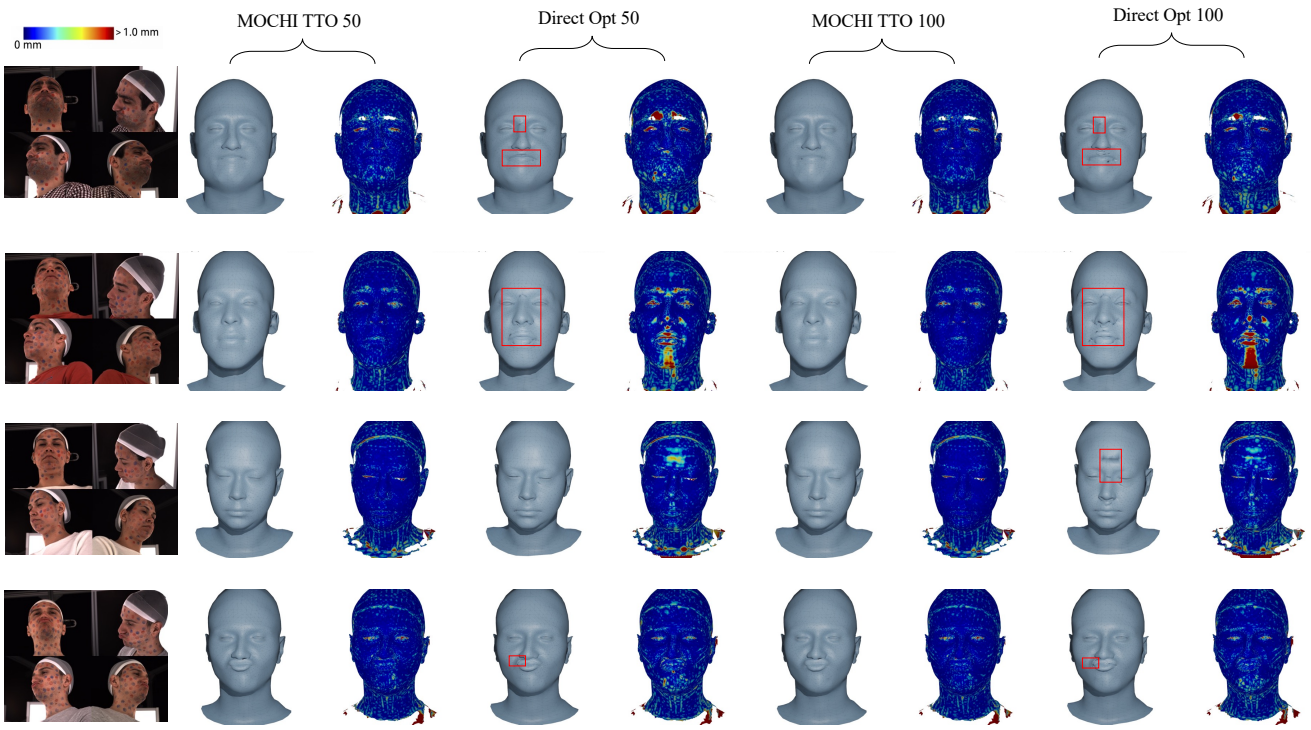

图12：优化策略的鲁棒性。我们将潜在空间优化（MOCHI TTO）与直接优化网格顶点（Direct Opt）在50次和100次迭代下进行对比。虽然直接优化降低了数值上的点到表面误差，但容易产生伪影、尖峰和表面不规则（以红框标出）。相比之下，MOCHI TTO利用网络学习到的先验知识，确保重建结果保持平滑且在解剖学上合理，同时精确拟合目标扫描。

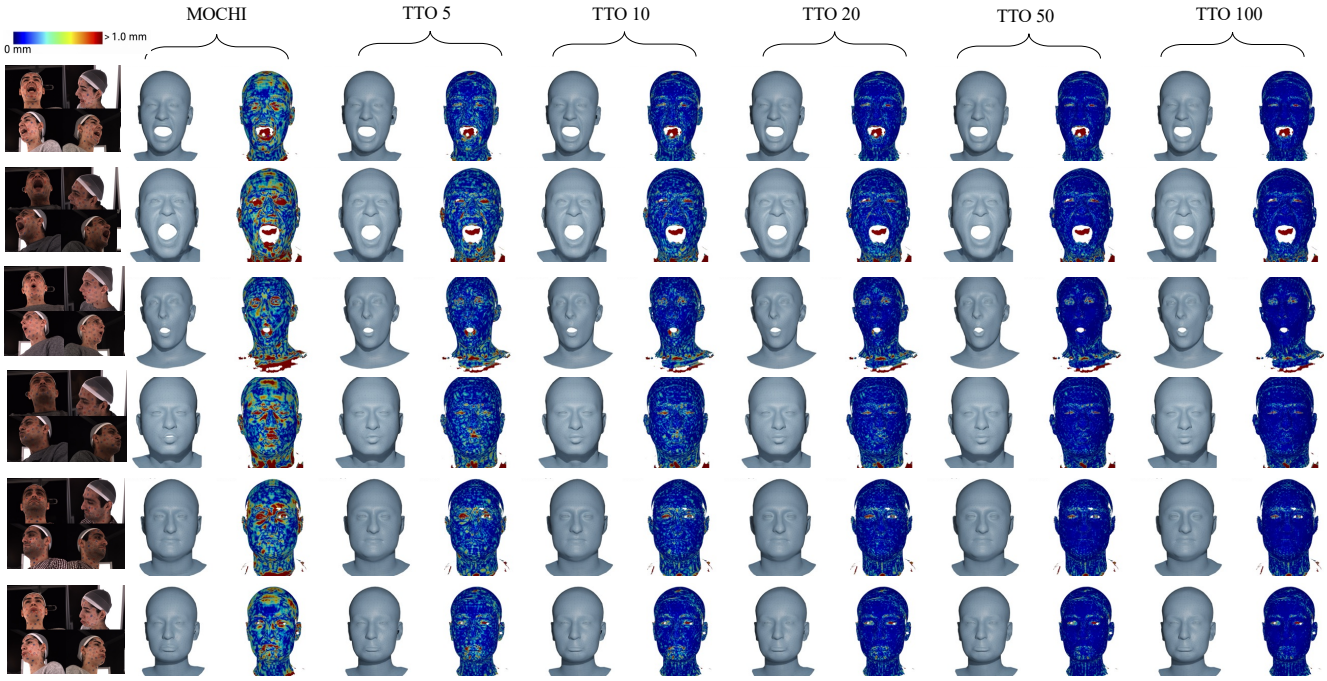

图13. 测试时优化的定性进展。该图展示了FaMoS测试集示例图像上网格细化在100次迭代中的演变过程。

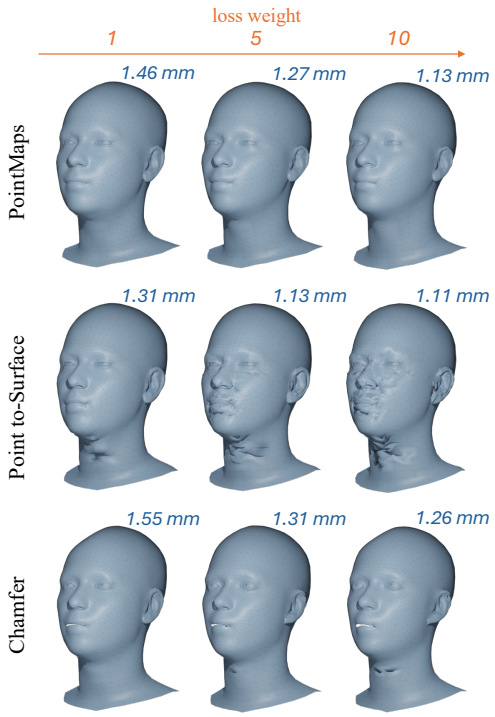

Figure 14. **Impact of loss type and weight.** We visualize the training of the coarse model using point maps (top), point-to-surface loss (middle), and Chamfer loss (bottom). When starting from a coarse prediction initialized only with landmark pretraining, increasing the point-to-surface loss weight leads to severe artifacts. While Chamfer distance results in fewer artifacts, it can still introduce errors (e.g., in the ears); in contrast, point-map supervision remains stable and topology-consistent. Blue numbers indicate the point-to-surface error (mm) relative to the ground-truth scan.

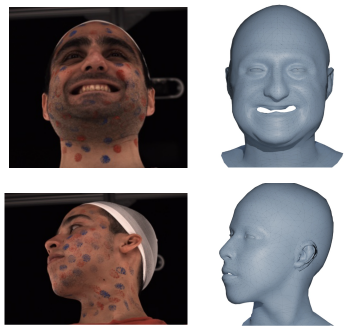

Figure 15. **Failure cases. Top Row (MOCHI TTO):** Since the FLAME topology lacks explicit teeth geometry, test-time optimization may incorrectly distort the lips by pulling them inward to minimize the point-to-surface distance to the teeth present in the raw scan. **Bottom Row (MOCHI):** In cases of large head poses, the base model may occasionally present with artifacts.

(1) fitting 3D morphable models (3DDMs) to in-the-wild images, which is limited by the fact that unconstrained face reconstruction is still an open problem;

(2) deriving landmarks from registrations of high-quality 3D face scans captured in controlled studio environments, which we explicitly aim to avoid, as our goal is not to depend on registrations;

(3) synthetically generating data, which offers full control over the facial geometry, appearance, and deformation, and therefore provides perfect ground truth.

Given these considerations, we adopt the third option.

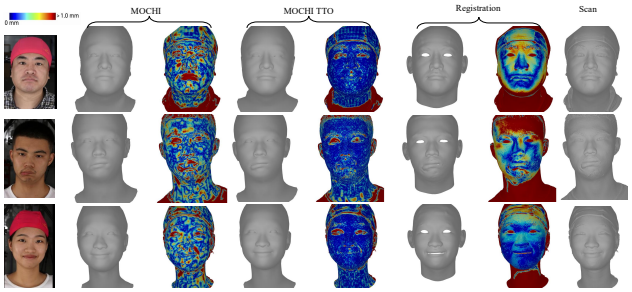

Figure 16. **Qualitative results on FaceScape.** From left to right: *MOCHI*, *MOCHI TTO*, the official *FaceScape registration*, and the *Scan*. For each method we show the predicted mesh (left) and the point-to-surface error heatmap (right; red $\geq$ 1.0 mm, lower is better).

Generating Geometry. To generate the geometry we randomly sample a FLAME identity, a FLAME expression, FLAME jaw opening angle, eyeball rotation and the activation of a separate eye-blinking blendshapes. To generate hair, we use HAAR [72] to sample 150 various hairstyles. For each generated face, we either keep it bald, or sample one of the hairstyles.

Albedo. We sample 54 (26 female and 28 male) very high quality textures purchased from 3D Scan Store, which cover all different skin tones and eye colors. These textures were re-topologized to be compatible with the FLAME UV-map. For each rendering, we uniformly sample one texture.

Illumination. For illumination, we employ high quality HDRI environment maps obtained from PolyHaven. They were captured in a variety of indoor and outdoor scenes and cover many possible illumination settings. For each rendering, we first sample one of 665 environment maps. Then, we randomly rotate the environment map about the vertical axis.

Camera position. We randomly sample the camera position to be in the frontal hemisphere of the rendered face. We also slightly vary the "look-at" point such that is not always

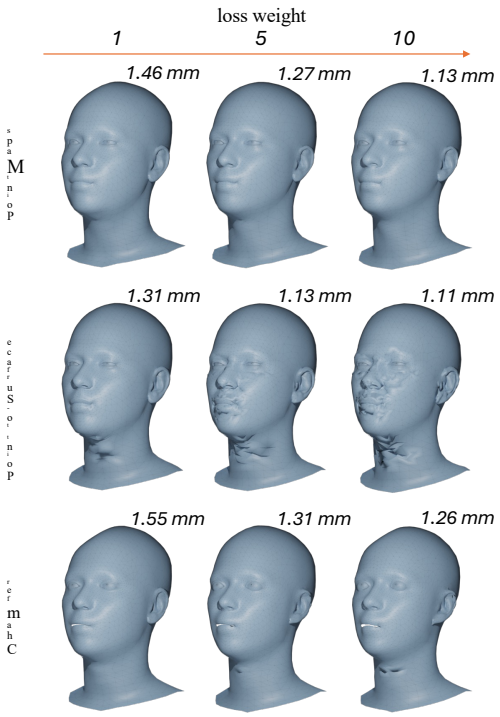

图14. 损失类型与权重的影响。我们使用点图（上）、点到面损失（中）和Chamfer损失（下）对粗模型训练进行可视化。当从仅以地标预训练初始化的粗预测开始时，增加点到面损失权重会导致严重伪影。虽然Chamfer距离产生的伪影较少，但仍可能引入误差（例如在耳朵处）；相比之下，点图监督保持稳定且拓扑一致。蓝色数字表示相对于真实扫描的点到面误差（毫米）。

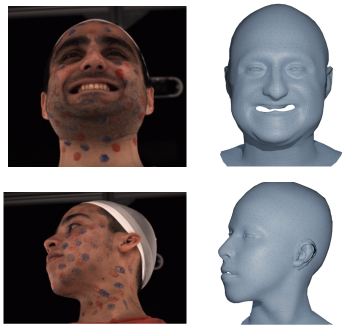

图15. 失败案例。顶行（MOCHI TTO）：由于FLAME拓扑缺乏显式牙齿几何结构，测试时优化可能错误地通过向内拉动嘴唇来最小化与原始扫描中牙齿的点到表面距离，从而扭曲嘴唇。底行（MOCHI）：在头部大姿态的情况下，基础模型偶尔会出现伪影。

(1) fitting 3D morphable models (3DDMs) to in-the-wild

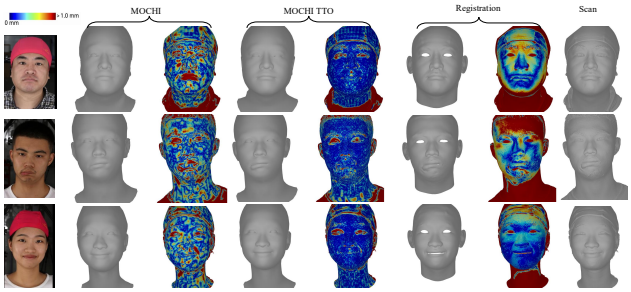

图16. FaceScape上的定性结果。从左到右依次为：*MOCHI*、*MOCHI TTO*、官方*FaceScape registration*和*Scan*。对于每种方法，我们展示了预测网格（左）和点到表面误差热图（右；红色≥ 1.0毫米，数值越低越好）。

图像，这受到无约束人脸重建仍是一个开放问题的限制；

(2) 从受控影棚环境中采集的高质量三维人脸扫描的配准中推导地标，这是我们明确希望避免的，因为我们的目标不是依赖配准；

(3) 合成生成数据，能够完全控制面部几何、外观和形变，因此提供了完美的真值。

鉴于这些考虑，我们采用第三种方案。

生成几何体。为了生成几何体，我们随机采样一个FLAME身份、一个FLAME表情、FLAME下颌张开角度、眼球旋转以及一个独立的眨眼混合形状的激活。为了生成头发，我们使用HAAR [72]采样150种不同的发型。对于每个生成的面部，我们要么保持光头，要么采样其中一种发型。

反照率。我们从3D Scan Store购买了54个（26个女性，28个男性）非常高质量的纹理，这些纹理覆盖了所有不同的肤色和眼睛颜色。这些纹理经过重新拓扑，以与FLAME UV贴图兼容。对于每次渲染，我们统一采样一个纹理。

光照。对于光照，我们采用从PolyHaven获取的高质量HDRI环境贴图。这些贴图在多种室内和室外场景中拍摄，覆盖了许多可能的光照设置。对于每次渲染，我们首先从665张环境贴图中采样一张。然后，我们随机绕垂直轴旋转环境贴图。

相机位置。我们随机采样相机位置，使其位于渲染人脸的前半球。我们还轻微变化“注视”点，使其不总是

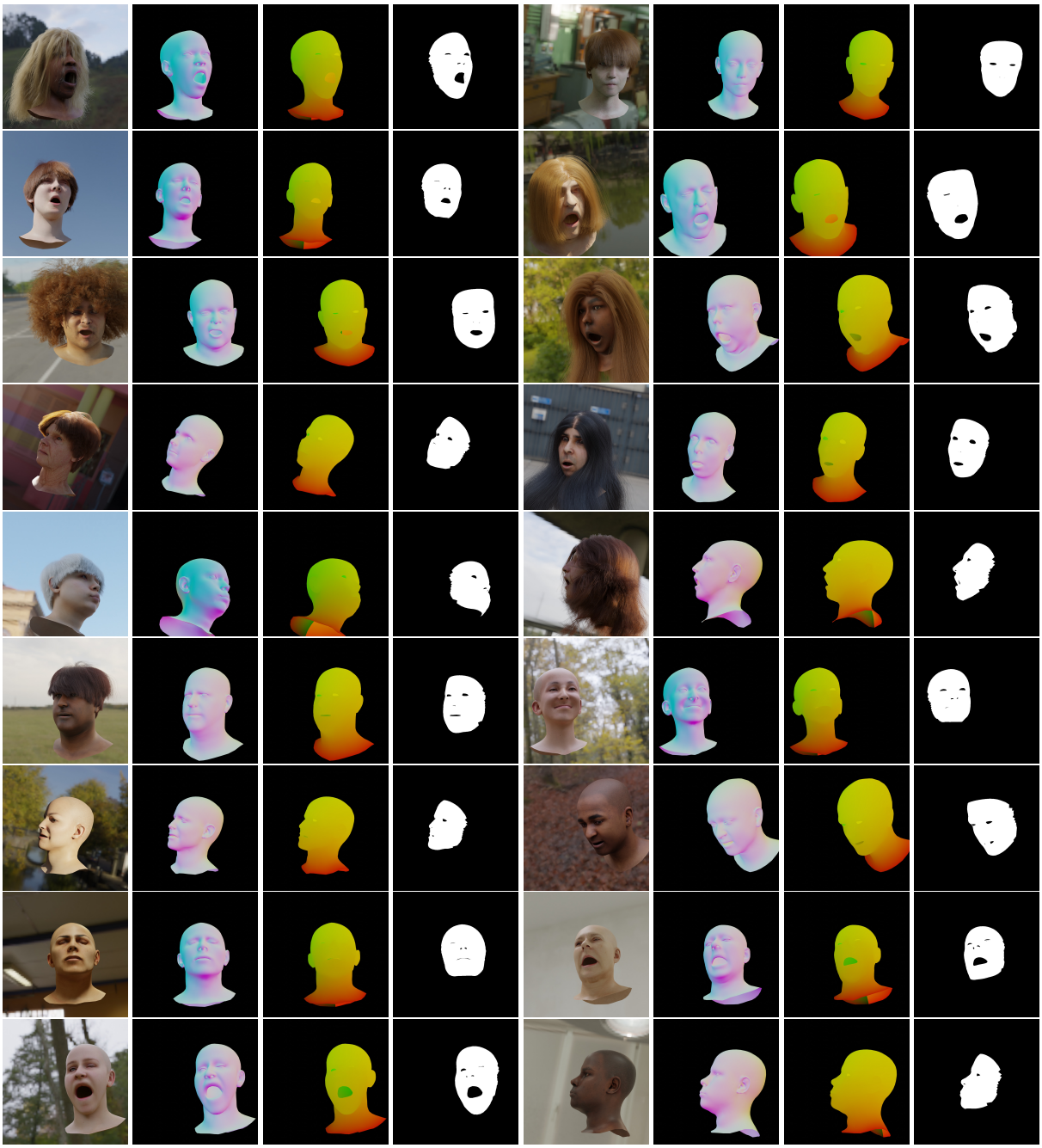

Figure 17. **Synthetic data.** From left to right: Path-traced RGB image, normal map, uv-map, binary face mask. We encourage the reader to zoom in on this figure in its digital form.

looking at the center of the face. Furthermore, we perturb the camera distance and the focal length.

Output. We use Blender [16] to render all the images. First, the Cycles path tracer renders the RGB image of the scene generated with the process described above. Then we remove hair (if the sample had any) and switch to the EEVEE renderer to render UV-map, normal map, and flame regions masks. The resulting images along with the mesh and camera parameters, provide perfect GT to train our dense landmark tracker. Figure 17 shows a few example outputs.

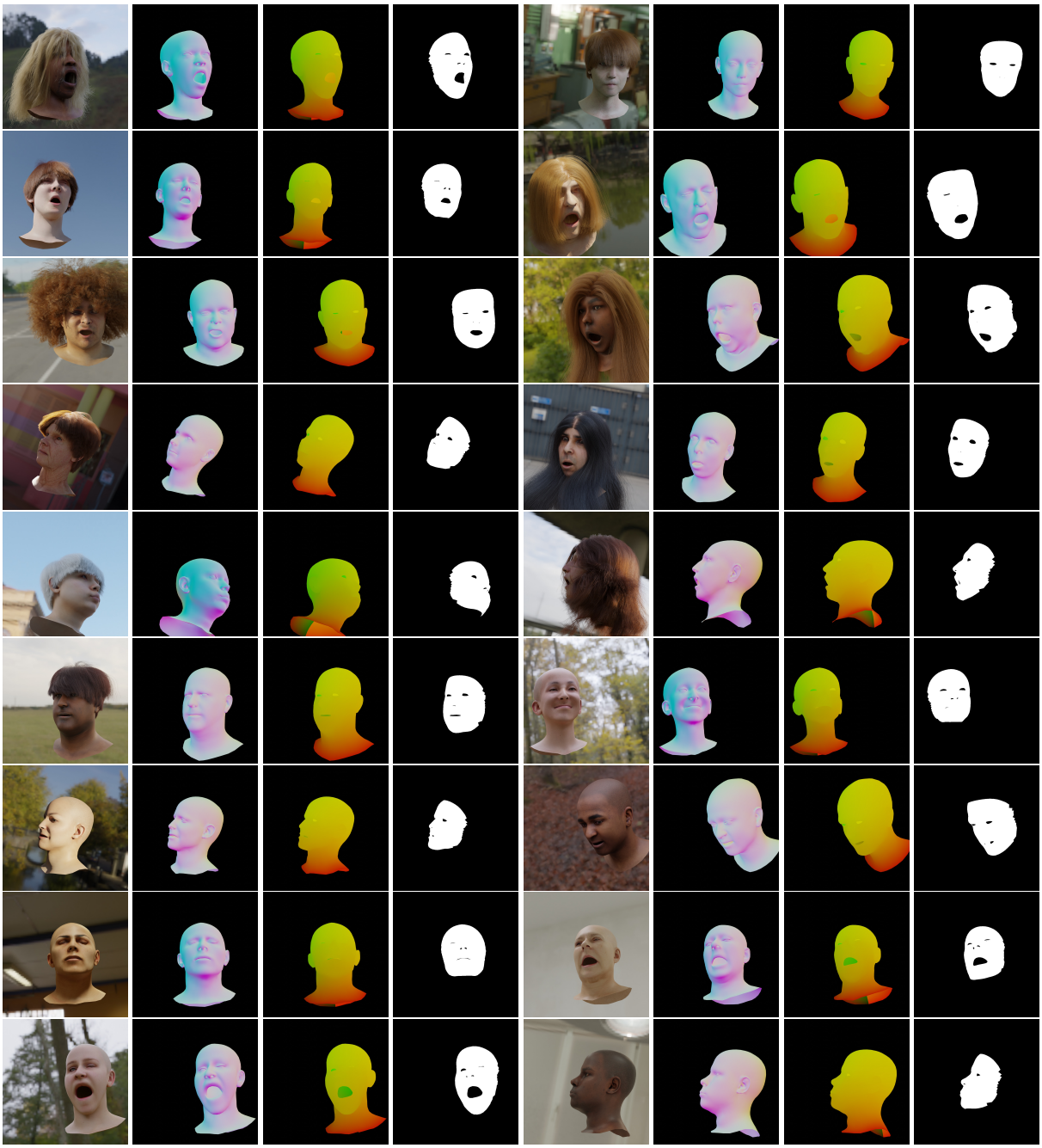

图17. 合成数据。从左到右：路径追踪RGB图像、法线贴图、UV贴图、二值人脸掩码。我们鼓励读者 er
要放大此图的数字版本。

注视面部中心。此外，我们扰动相机距离和焦距。

输出。我们使用Blender [16]渲染所有图像。首先，Cycles路径追踪器渲染通过上述过程生成的场景的RGB图像。然后我们移除头发（如果样本有的话）并切换到EEVEE渲染器用于渲染UV贴图、法线贴图和火焰区域掩码。生成的图像连同网格和相机参数，为训练我们的密集地标跟踪器提供了完美的真值。图17展示了一些示例输出。

Limitations. While our data is sufficient for our tasks, it does have a few limitations. First, FLAME models the face and neck only, which results in a floating-head appearance. Although this is not realistic, we do not observe any negative impact on model performance. Second, we do not model the mouth cavity, teeth, or tongue, as these components are not included in FLAME. Finally, our synthetic data does not include facial hair or accessories such as glasses, which may lead to reduced performance when these are present. Nevertheless, for our primary use case, namely landmark tracking for multi-view facial capture, subjects typically do not exhibit such variations, so this limitation has minimal practical impact.

D.2. 2D Landmark Tracker

Using the previously described synthetic dataset, we train a dedicated 2D landmark tracker. We employ a Vision Transformer (ViT-Large) backbone [19], initialized with DINOv3 weights [71]. To maintain the robust semantic features learned during pre-training while adapting to our task, we freeze the backbone parameters and inject Low-Rank Adaptation (LoRA) [37] modules into the attention and feed-forward blocks. Specifically, we apply LoRA with rank $r = 8$ and $\alpha = 16$ to the query, key, value, and projection matrices in the attention layers, as well as the fully connected layers in the MLP blocks. The features are aggregated via global average pooling and passed to a lightweight regression head consisting of a LayerNorm and a final Linear projection that predicts the 2D normalized coordinates of the FLAME mesh vertices.

Training and Evaluation. We train the tracker for 100 epochs on our generated synthetic dataset. The input images are resized to 512×512 pixels. We use the AdamW optimizer [54] with a learning rate of 10^{-4} and weight decay of 10^{-4}. The learning rate is scheduled using a cosine annealing strategy with a linear warm-up. The model is supervised using a Mean Squared Error (MSE) loss between the predicted 2D coordinates and the ground-truth projected vertices from the synthetic rendering. To improve generalization to real-world data, we apply aggressive geometric and photometric image augmentations during training.

To validate our 2D tracker, we perform a qualitative comparison against projected FLAME registrations (from a classical pipeline) on a subset of the FaMoS dataset. Figure 18 visualizes the output of our tracker—trained exclusively on synthetic data—when applied to real-world samples. In our MOCHI pipeline, these 2D predictions serve as semantic regularizers, anchoring difficult regions such as the eyes and lips. As such, the tracker does not need to achieve pixel-perfect accuracy, but rather provide stable, semantically meaningful guidance.

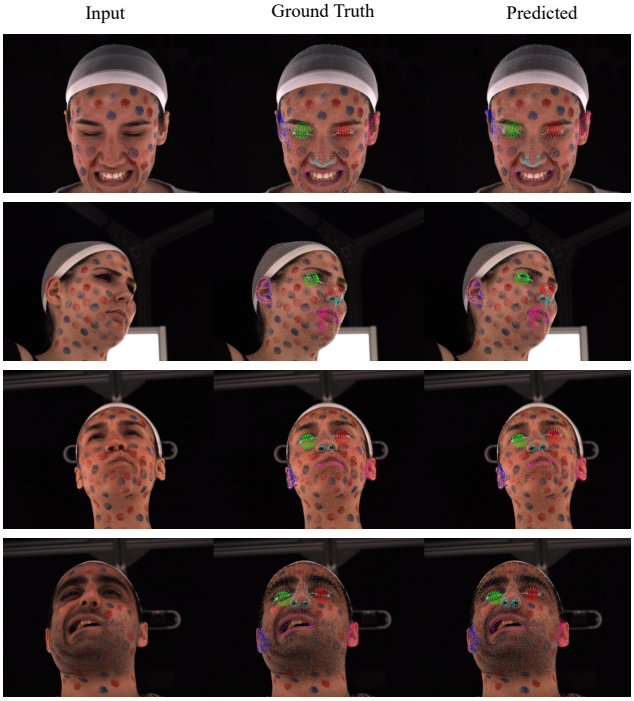

Figure 18. **Qualitative evaluation of the 2D landmark tracker on the FaMoS dataset.** From left to right: the input RGB image, the Ground Truth landmarks (obtained by projecting the available registration), and the landmarks predicted by our tracker. The vertices are color-coded by semantic region (e.g., green/red for eyes, pink for lips, blue for ears). Despite being trained exclusively on synthetic data, the tracker generalizes robustly to real-world images, accurately capturing dense correspondence across challenging expressions and poses.

局限性。尽管我们的数据足以完成相关任务，但仍存在一些局限。首先，FLAME仅对面部和颈部进行建模，导致出现“漂浮头部”的外观。虽然这并不真实，但我们未观察到其对模型性能产生负面影响。其次，我们未对口腔、牙齿或舌头进行建模，因为这些组成部分未包含在FLAME中。最后，我们的合成数据未包含面部毛发或眼镜等配饰，这可能导致在存在这些元素时性能下降。尽管如此，对于我们的主要应用场景，即多视角面部捕捉中的地标跟踪，受试者通常不会表现出此类变化，因此这一局限的实际影响极小。

D.2. 二维地标跟踪器

利用前述合成数据集，我们训练了一个专用的2D地标跟踪器。我们采用Vision Transformer（ViT-Large）骨干网络[19]，并使用DINOv3权重[71]进行初始化。为在保持预训练期间学到的鲁棒语义特征的同时适应我们的任务，我们冻结骨干网络参数，并在注意力模块和前馈模块中注入低秩自适应（LoRA）[37]模块。具体而言，我们对注意力层中的查询、键、值及投影矩阵，以及MLP块中的全连接层应用秩为$r = 8$和$\alpha = 16$的LoRA。特征通过全局平均池化进行聚合，并传递给一个轻量级回归头，该回归头由LayerNorm和最终线性投影组成，用于预测FLAME网格顶点的2D归一化坐标。

训练与评估。我们在生成的合成数据集上对跟踪器训练100个epoch。输入图像被调整为512×512像素。我们使用AdamW优化器[54]，学习率为10^{-4}，权重衰减为10^{-4}。学习率采用余弦退火策略并配合线性预热进行调度。模型通过预测的2D坐标与合成渲染中真实投影顶点之间的均方误差（MSE）损失进行监督。为提高对真实世界数据的泛化能力，我们在训练期间应用了激进的几何和光度图像增强。

为了验证我们的2D跟踪器，我们在FaMoS数据集的一个子集上，与经典流程投影得到的FLAME配准结果进行了定性比较。图18展示了我们的跟踪器——仅使用合成数据训练——应用于真实世界样本时的输出。在我们的MOCHI流程中，这些2D预测作为语义正则化器，锚定眼睛和嘴唇等困难区域。因此，跟踪器无需达到像素级完美精度，而只需提供稳定、具有语义意义的引导。

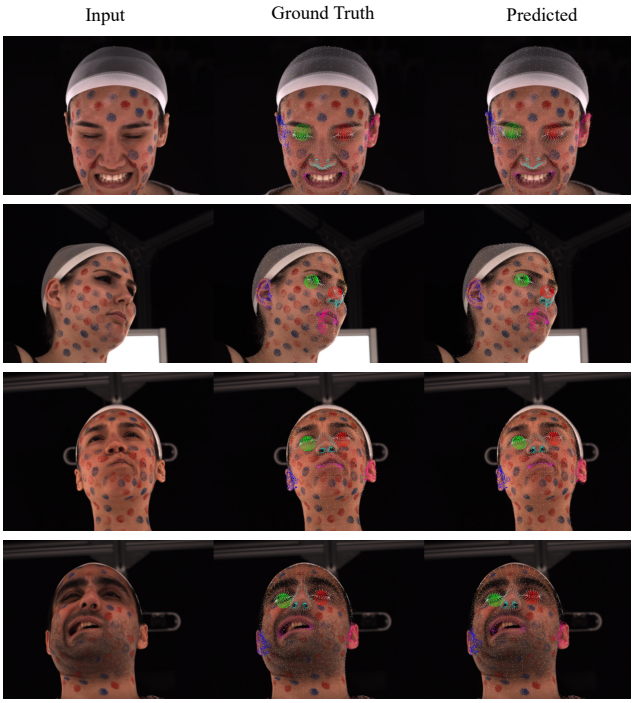

图18. 在FaMoS数据集上对2D地标跟踪器的定性评估。从左到右依次为：输入RGB图像、真实地标（通过投影可用配准获得）以及我们的跟踪器预测的地标。顶点按语义区域进行颜色编码（例如，绿色/红色表示眼睛，粉色表示嘴唇，蓝色表示耳朵）。尽管仅使用合成数据进行训练，该跟踪器仍能稳健地泛化到真实世界图像，在具有挑战性的表情和姿态下准确捕捉密集对应关系。